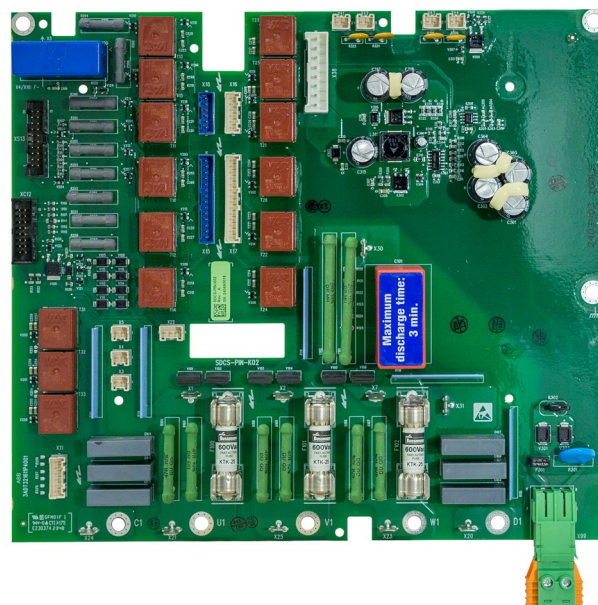
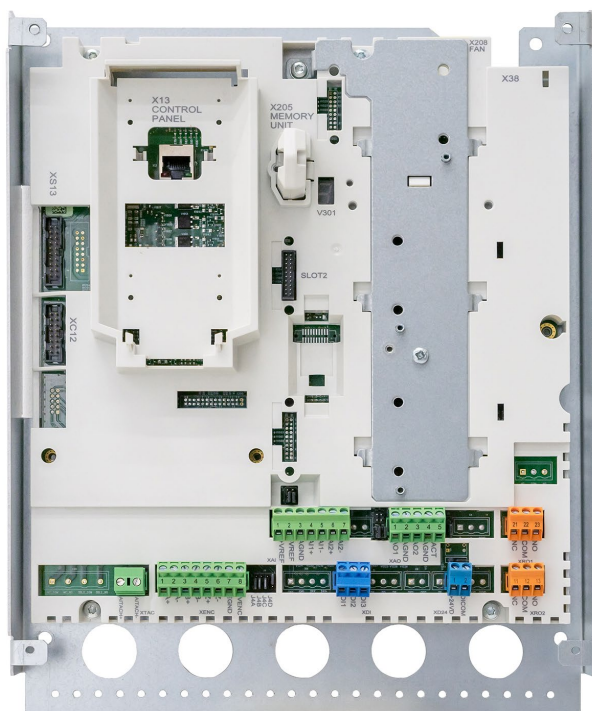


DC DRIVES

ABB general purpose drives

DCS580 Drives - Service manual



DCS580 Drive Manuals

Name	Publication number	EN	DE	IT	ES	FR	PL	ZH	RU	JP
DCS580 Quick guide	3ADW000827	EN								
DCS580 Drives										
DCS580 Flyer	3ADW000859	EN								
DCS580 Technical catalog	3ADW000860	EN								
DCS580 Hardware manual	3ADW000826	EN								
DCS580 Firmware manual	3ADW000820	EN								
DCS580 Service manual	3ADW000865	EN								
ACS-AP-x assistant control panels user's manual	3AUA0000085685	EN								
DCS Thyristor power converter – Technical guide	3ADW000163	EN								
Door mounting kits										
DPMP-01 mounting platform for ACS-AP control panel	3AUA0000100140	EN								
DPMP-02 mounting platform for ACS-AP control panel	3AUA0000136205	EN								
Serial communication										
FPBA-01 PROFIBUS DP adapter module	3AFE68573271	EN	DE				PL	ZH		
FEIP-21 EtherNet/IP fieldbus adapter module	3AXD50000158621	EN								
FMBT-21 Modbus/TCP adapter module	3AXD50000158607	EN								
FPNO-21 PROFINET fieldbus adapter module	3AXD50000158614	EN								
Tool and maintenance manuals and guides										
Drive Composer PC tool	3AUA0000094606	EN								
Adaptive programming, Application guide	3AXD50000028574	EN								

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Safety instructions

What this chapter contains

This chapter contains the safety instructions you must follow when installing, operating, and servicing the drive.

If ignored, physical injury or death may follow, or damage may occur to the drive, the motor or driven equipment. Read the safety instructions before you work on the drive.

To which products this chapter applies

The information is valid for the entire range of the product DCS580, the converter modules DCS580-S0x size K1 ... K4.

Usage of warnings and notes

There are two types of safety instructions throughout this manual: Warnings and notes. Warnings caution you about conditions which can result in severe injury or death and/or damage to the equipment, and advice on how to avoid the danger. Notes draw attention to a particular condition or fact or give information on a subject.

The warning symbols are used as follows:



Dangerous voltage warning warns of high voltage which can cause physical injury or death and/or damage to the equipment.



General danger warning warns about conditions, other than those caused by electricity, which can result in physical injury or death and/or damage to the equipment.



Electrostatic sensitive devices warning warns of electrostatic discharge which can damage the equipment.

Installation and maintenance work

These warnings are intended for all who work on the drive, motor cable or motor. Ignoring the instructions can cause physical injury or death and/or damage to the equipment.



WARNING

- **Only qualified electricians are allowed to install and maintain the drive!**
- Never work on the drive, motor cable, or motor when main power is applied.
- Always ensure by measuring with a multimeter (impedance at least 1 MΩ) that:
 1. The voltage between the drive mains phases U1, V1 and W1 and the frame is close to 0 V.
 2. The voltage between terminals C(+) and D(-) and the frame is close to 0 V.
- Do not work on the control cables when power is applied to the drive or to the external control circuits. Externally supplied control circuits may cause dangerous voltages inside the drive even when the mains power on the drive is switched off.
- Do not make any insulation resistance or voltage withstand tests on the drive.
- Isolate the motor cables from the drive when testing the insulation resistance or voltage withstand of the cables or the motor.
- When reconnecting the motor cable, always check that the C(+) and D(-) cables are connected with the proper terminal.

Notes:

- The motor cable terminals on the drive are at a dangerously high voltage when the mains power is on, regardless of whether the motor is running or not.
- Depending on the external wiring, dangerous voltages (115 V, 220 V or 230 V) may be present on the relay outputs of the drive system (e.g., XRO1 ... XRO3).

- DCS580 with enclosure extension: Before working on the drive, isolate the whole drive system from the supply.

Grounding

These instructions are intended for all who are responsible for the grounding of the drive. Incorrect grounding can cause physical injury, death and/or equipment malfunction and increase electromagnetic interference.



WARNING

- Ground the drive, motor, and adjoining equipment to ensure personnel safety in all circumstances, and to reduce electromagnetic emission and pick-up.
- Make sure that grounding conductors are adequately sized and marked as required by safety regulations.
- In a multiple-drive installation, connect each drive separately to protective earth (PE).
- Minimize EMC emission and make a 360° high frequency grounding of screened cable entries at the cabinet lead-through plate.
- Do not install a drive equipped with an EMC filter to an ungrounded power system or a high resistance-grounded ($> 30 \Omega$) power system.

Notes:

- Power cable shields are suitable as equipment grounding conductors only when adequately sized to meet safety regulations.
- As the normal leakage current of the drive is higher than 3.5 mA_{AC} or 10 mA_{DC} a fixed protective earth connection is required.
- This product can cause a DC current in the protective earthing conductor. Where a residual current-operated protective (RCD) or monitoring (RCM) device is used for protection in case of direct or indirect contact, only an RCD or RCM of Type B is allowed on the supply side of this product.

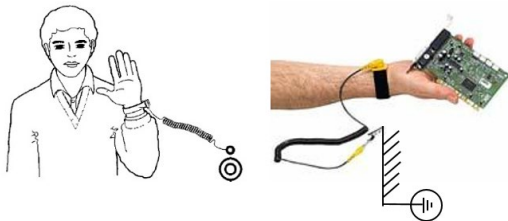
Printed circuit boards

These instructions are intended for all who handle the circuit boards. Ignoring the following instructions can cause damage to the equipment.



WARNING

- The printed circuit boards contain components sensitive to electrostatic discharge. Wear a grounding wristband when handling the boards. Touch the boards only when necessary.
- Grounding wristband:



- ABB order no.: 3ADV050035P0001.

Mechanical installation

These notes are intended for all who install the drive. Handle the unit carefully to avoid damage and injury.



WARNING

- DCS580 size K4:
 - The drive is heavy. Lift the drive by lifting lugs only.
 - The drive's center of gravity is high. Do not tilt it. It will overturn from a tilt of about 6 degrees. An overturning drive can cause physical injury.
 - Do not lift the drive by the front cover.
 - Place drives K4 only on their back.

- Make sure that dust from drilling does not enter the drive when installing. Electrically conductive dust inside the unit may cause damage or lead to malfunction.
- Ensure sufficient cooling.
- Do not fasten the drive by riveting or welding.

Operation

These warnings are intended for all who plan the operation of the drive or operate the drive. Ignoring the instructions can cause physical injury or death and/or damage to the equipment.



WARNING

- Before adjusting the drive and putting it into service, make sure that the motor and all driven equipment are suitable for operation throughout the speed range provided by the drive. The drive can be adjusted to operate the motor at speeds above and below the base speed.
- Do not control the motor with the load disconnecting device (disconnecting mains); instead, use the control panel keys  and , or commands via the I/O board of the drive.
- Mains connection:
You can use a disconnect switch (with fuses) to disconnect the electrical components of the drive from the mains for installation and maintenance work. The type of disconnect switch used must be as per EN 60947-3, Class B, so as to comply with EU regulations, or a circuit-breaker type which switches off the load circuit by means of an auxiliary contact causing the breaker's main contacts to open. The mains disconnect must be locked in its "OPEN" position during any installation and maintenance work.
- EMERGENCY STOP buttons must be installed at each control desk and at all other control panels requiring an emergency stop function. Pressing the STOP button on the control panel of the drive will neither cause an emergency stop of the motor, nor will the drive be disconnected from any dangerous potential.
- To avoid unintentional operating states, or to shut the unit down in case of any imminent danger according to the standards in the safety instructions it is not sufficient to merely shut down the drive via signals "RUN", "OFF" or "Emergency Stop" respectively "control panel" or "PC tool".
- Intended use:
The operating instructions cannot take into consideration every possible case of configuration, operation, or maintenance. Thus, they mainly give such advice only, which is required by qualified personnel for normal operation of the machines and devices in industrial installations.
If in special cases the electrical machines and devices are intended for use in non-industrial installations - which may require stricter safety regulations (e.g., protection against contact by children or similar) - these additional safety measures for the installation must be provided by the customer during assembly.

Note:

- When the control location is not set to Local (Local not shown in the status row of the display), the stop key on the control panel will not stop the drive. To stop the drive using the control panel, press the Loc/Rem key and then the stop key .

Disclaimers

Generic disclaimers

The manufacturer shall have no obligation with respect to any product which:

- (i) Has been improperly repaired or altered.
- (ii) Has been subjected to misuse, negligence, or accident.
- (iii) Has been used in a manner contrary to the manufacturer's instructions.
- (iv) Has failed as a result of ordinary wear and tear.

The information in this document is subject to change without notice and should not be construed as a commitment by ABB.

Cybersecurity disclaimer

This product is designed to be connected to and to communicate information and data via a network interface. The HTTP protocol, which is used between PC tool (Drive Composer) and product, is an unsecured protocol. For independent and continuous operation of the product such a connection via network to a commissioning tool is not necessary. However, it is the customer's sole responsibility to provide and continuously ensure a secure connection between the product and the customer network or any other network (as the case may be). The customer shall establish and maintain appropriate measures (such as but not limited to the installation of firewalls, prevention of physical access, application of authentication measures, encryption of data, installation of anti-virus programs, etc.) to protect the product, the network, its system and the interface against any kind of security breaches, unauthorized access, interference, intrusion, leakage and/or theft of data or information.

Notwithstanding any other provision to the contrary and regardless, whether the contract is terminated or not, ABB and its affiliates are under no circumstances liable for damages and/or losses related to such security breaches, any unauthorized access, interference, intrusion, leakage and/or theft of data or information.

Introduction to this manual

Chapter overview

This chapter describes the purpose, contents, and the intended use of this manual.

Before You Start

The purpose of this service manual is to provide detailed information on how to service DCS580 power converters. The [Safety instructions](#) at the beginning of this manual needs to be studied before attempting any work on or with the drive. Read this manual before servicing the drive.

What this manual contains

The [Safety instructions](#) can be found at the beginning of this manual.

[Introduction to this manual](#), the chapter you are currently reading, introduces you to this manual.

[Fault Tracing Thyristors](#), this chapter describes how to detect a faulty thyristor or SDCS-BAB-F0x.

[Handling the semiconductors](#), this chapter describes the handling of thyristor modules.

[Exchange thyristors sizes K1 ... K4](#), this chapter describes the exchange of thyristors and SDCS-BAB-F0x in drives size K1 ... K4.

[Exchange the electronic tray \(SDCS-CON-K02L\)](#), this chapter describes the exchange of an electronic tray containing the SDCS-CON-K02L.

[Service](#), this chapter contains hardware change information, firmware download and technical hints.

[Preventive Maintenance](#), this chapter describes preventive maintenance of the drives.

[Appendix: Spare parts](#), the Appendix contains the spare parts information.

Target group

This manual is designed to help those responsible for planning, installing, starting up and servicing drives. These people should possess:

- Basic knowledge of physics, electrical engineering, electrical wiring principles, components as well as symbols used in electrical engineering.
- Basic experience with DC drives and DC products.

Related documents

A list of related manuals is shown on the inside of the front cover under [DCS580 Drive Manuals](#).

Storage and transport

If the drive has been in storage prior to installation or is transported to another location, care must be taken to ensure that the environmental conditions are complied with, see [DCS580 Hardware Manual \(3ADW000826\)](#).

Terms and abbreviations

Term/Abbreviation	Definition
AC 800M	A type of programmable controller manufactured by ABB.
ACS-AP-I	Types of control panels.
ACS-AP-W	
Adaptive Program (AP)	Adaptive Program. See Adaptive programming, Application guide (3AXD50000028574)
AI	Analog input. Interface for analog input signals.
AO	Analog output. Interface for analog output signals.
Application program	Program written by the Drive Application Builder. See Drive (IEC61131-3) application programming manual (3AUA0000127808) .
Control unit	It contains electronics and I/O connections. The control unit is connected to the power unit.
Converter	Converter to control DC motors/loads.
D2D	Drive-to-drive/Device-to-device link. It is a communication link between ABB equipment.
DCS580	A product family of ABB drives.
DCS880	A product family of ABB drives.
DCSLink	Communication between the armature converter and the field exciters or 12-pulse communication.
DCT880	A product family of ABB drives (thyristor power controller).
DDCS	Distributed Drives Communication System. A protocol used in communication between ABB equipment.
DI	Digital input. Interface for digital input signals.
DIO	Digital input/output. Interface that can be used as a digital input or output.
DO	Digital output. Interface for digital output signals.
Drive	Converter to control DC motors.
DriveBus	A communication link used between ABB equipment.
DriveAP	Adaptive Programming. See Adaptive programming, Application guide (3AXD50000028574) .
Drive Application Builder	Tool to write application programs. See Drive (IEC61131-3) application programming manual (3AUA0000127808) .
Drive Composer	PC tool for commissioning and maintenance of ABB equipment.
EFB	Embedded fieldbus.
FAIO-01	Optional analog I/O extension module.
FBA	Fieldbus adapter.
FCAN-01	Optional CANOpen adapter.
FCNA-01	Optional ControlNet adapter.
FDCO-0x	Optional DDCS communication module.
FDIO-01	Optional digital I/O extension module.
FDNA-01	Optional DeviceNet adapter.
FEA-03	Optional I/O extension module.
FECA-01	Optional EtherCAT® adapter.
FEIP-21	Optional EtherNet/IP fieldbus adapter module.
FEN-01	Optional TTL encoder interface module.
FEN-11	Optional absolute encoder interface module.
FEN-21	Optional resolver interface module.
FEN-31	Optional HTL encoder interface module.
FENA-11	Optional Ethernet/IP, Modbus/TCP and PROFINET IO adapter.
FENA-21	Optional dual-port Ethernet/IP, Modbus/TCP, and PROFINET IO adapter.
FEPL-02	Optional POWERLINK adapter.
FIO-01	Optional digital I/O extension module.

Term/Abbreviation	Definition
FIO-11	Optional analog I/O extension module.
FMBT-21	Optional Modbus/TCP adapter module.
FPBA-01	Optional PROFIBUS DP adapter.
FPNO-21	Optional PROFINET fieldbus adapter module.
FSCA-01	Optional Modbus/RTU adapter.
FSE-31	Optional HTL safety encoder interface module.
FSO-21	Optional safety functions module.
FSPS-21	Optional PROFIsafe safety functions module.
HTL	High-threshold logic.
I/O	Input/Output.
ModuleBus	A communication link used between ABB equipment. Can be connected to the optical ModuleBus link.
Network control	With fieldbus protocols based on the Common Industrial Protocol (CIP™), such as DeviceNet and Ethernet/IP, denotes the control of the connected equipment using the Net Ctrl and Net Ref objects of the ODVA AC/DC Drive Profile. For more information, see www.odva.org , and the following manuals: – FDNA-01 DeviceNet adapter module User's manual (3AFE68573360). – FENA-11/-21 Ethernet adapter module User's manual (3AUA0000093568).
Off3 (emergency stop)	Off3 (emergency stop) with configurable deceleration time according to cat. 1.
OPL	Optical power link. Protocol used in communication between the control unit and the power unit.
Parameter	User-adjustable operation instruction to the drive/converter.
PID controller	Proportional-integral-derivative controller. E.g., the speed/voltage/temperature control is based on a PID algorithm.
PLC	Programmable logic controller.
Power unit	It contains power electronics and power connections. The control unit is connected to the power unit.
PTC	Positive temperature coefficient.
PU	See power unit.
RFG	Ramp function generator.
RO	Relay output. Interface for a digital output signal. Implemented with a relay.
Signal	Value measured or calculated. It can also contain status information. Most signals are read-only, but some (especially counter-type signals) can be reset.
SS1	Safe stop 1.
SSI	Synchronous serial interface.
STO	Safe torque off.
TTL	Transistor-transistor logic.
UPS	Uninterruptible power supply. Power supply equipment with battery to maintain output voltage during power failure.
Unit	Thyristor power controller to control heating loads.

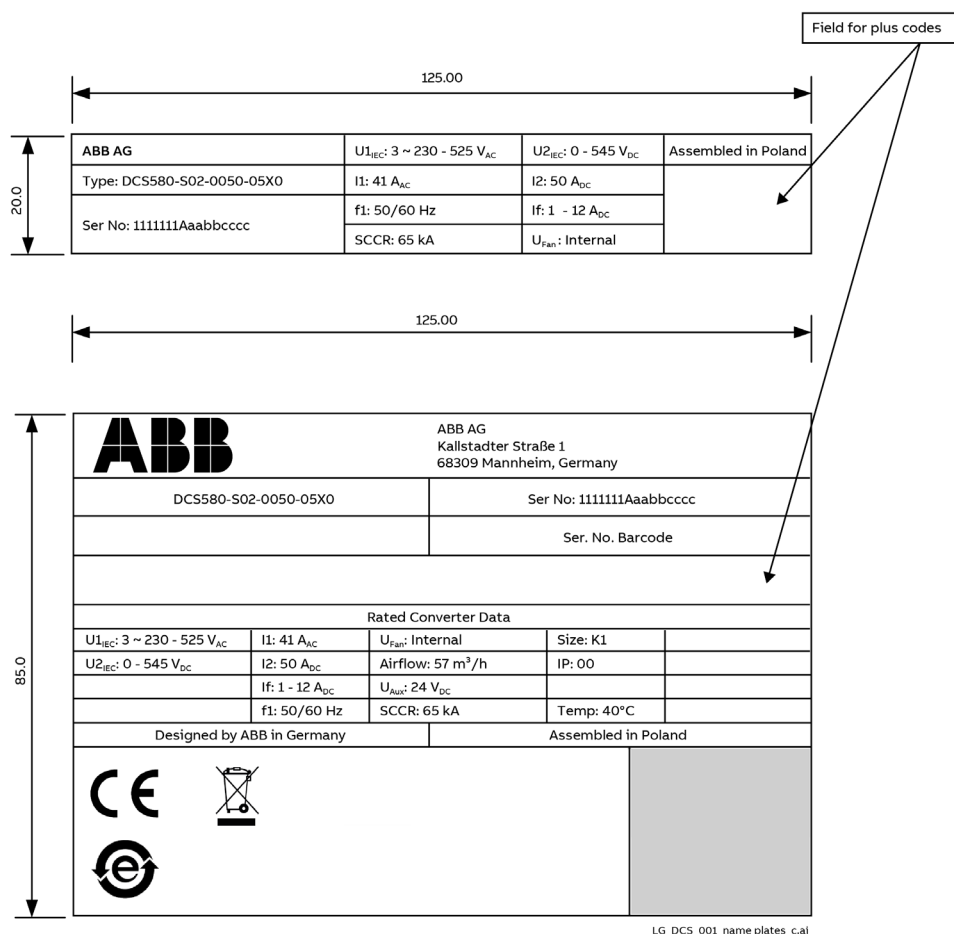
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Name plate

For purposes of identification, each drive is fitted with name plates, stating the type code and the serial number, which serve for each drive's individual identification.

The type code contains information about the characteristics and the configuration of the drive.



LG_DCS_001_name plates_c.ai

Ser. No.	E.g., 0025421A24044264. 2404 = Production year 2024 and week 04.
U _{1IEC}	Rated input voltage according to IEC.
U _{2IEC}	Rated output voltage according to IEC.
I ₁	Rated input current.
I ₂	Rated output current.
I _f	Rated internal field exciter current.
f ₁	Rated frequency of mains voltage.
U _{Fan}	Rated fan voltage.
Airflow	Rated cooling air flow.
U _{Aux}	Rated auxiliary voltage.
SCCR	Short circuit current ratio.
Size	Unit size.
IP: 00	Protection class according to ISO20653.
Temp	Max. permissible cooling air temperature.

Type code

The type code contains information on the specification and configuration of the drive. The first digits from the left show the basic configuration (e.g., DCS580-S01-1000). The optional selections are given thereafter on the name plate by plus code. The main selections are described below. Not all selections are available for all types.

The drive's basic type code: DCS580-aab-cccc-ddef			
Product family:	DCS580		
Product type:	aa	= S0	Standard converter module
Bridge type:	b	= 1	Single bridge (2-Q)
		= 2	2 anti-parallel bridges (4-Q)
Module type:	cccc	=	Rated DC current (IP00)
Rated AC voltage:	dd	= 05	230 V _{AC} ... 525 V _{AC}
Power connection:	e	= X	Standard K1 ... K4
Revision code:	f	= 0	1 st generation

The technical data and specifications are valid as of going to press. ABB reserves the right to make subsequent alterations.

Voltage ratings

Armature voltages

The maximum available armature voltages have been calculated using the following assumptions:

- U_{VN} = rated mains voltage, 3-phase.
- Voltage tolerance $\pm 10\%$.
- Internal voltage drop approximately 1 %.

If a deviation or a voltage drop has to be considered in compliance with IEC and VDE standards, the output voltage and/or the output current must be reduced.

Mains voltage	Maximum DC voltage		Ideal DC voltage	DC voltage class
U_{VN} [V _{AC}]	$U_{d \max 2-Q}$ [V _{DC}]	$U_{d \max 4-Q}$ [V _{DC}]	U_{d0} [V _{DC}]	
230	265	240	310	05
380	440	395	510	05
400	465	415	540	05
415	480	430	560	05
440	510	455	590	05
460	530	480	620	05
480	555	500	640	05
500	580	520	670	05
525	610	545	700	05

Field voltages

The maximum available field voltage can be calculated using the following formula:

$$U_F \leq 1.35 * U_{VN} * \left(\frac{100\% * TOL}{100\%} \right), \text{ with:}$$

U_F = Field voltage.

U_{VN} = Mains voltage.

TOL = Tolerance of the mains voltage in %.

Current ratings

Drive size	2-Q rated current DCS580-01 [A _{DC}]	4-Q rated current DCS580-02 [A _{DC}]	Supply voltage [V _{AC}]
			525
K1	20	25	X
	45	50	X
	65	75	X
	90	100	X
K2	135	150	X
	180	200	X
	225	250	X
	270	300	X
K3	315	350	X
	405	450	X
	470	520	X
K4	610	680	X
	740	820	X
	900	1000	X

Fault Tracing Thyristors

Tools

For commissioning and fault tracing

The following software tools are mandatory:

- Drive Composer **pro**. Includes commissioning wizard, Adaptive Program, and fast drive monitoring.

The following tools are mandatory in addition to standard tools:

- An oscilloscope including memory function with either galvanically isolating transformer or isolating amplifier (probe) for safe measurements. It can also be a handheld (portable) oscilloscope.
- A handheld tacho to measure the motor speed.



- A clamp on current probe. In case the scaling of the DC load current needs to be checked it must be a DC clamp on current probe.
- A voltmeter (at least CAT III 1000 V with possibility of capacitance measurement).



- 1000 V probes and test leads.



- An ESD-field service kit (ABB Service Finland code 0001ESD/MS-Antistatic).



- Make sure that all equipment in use is suitable for the voltage level applied to the power part!

Additionally for service and preventive maintenance:

The following additional tools are mandatory for cleaning:

- An ESD safe blower/ESD vacuum cleaner (ABB Service Finland code 0006ESD/MUNTZ 555-ESD-S-E).



How to detect a faulty thyristor

Thyristor problems can be noticed differently:

A fuse is blown

This is an indication that a strong overcurrent has happened due to one of the following reasons:

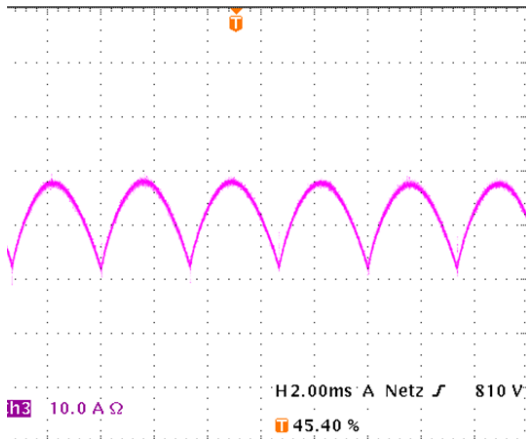
- An internal short circuit between the phases (line side/AC-side) because of a defective thyristor (short circuit inside a thyristor from anode to cathode).
- An internal short circuit between the phases (line side/AC-side) because of circulating current in a 4-Q drive (malfunction of the control electronics, no thyristor defective).
- An external short circuit at the DC terminals of the drive without sufficient impedance.
- A commutation fault during generating (active braking with high current, high EMF and with low AC voltage) of a 4-Q drive.

Notes:

- In case of parallel fuses: If one of the parallel fuses is blown, all parallel fuses have to be changed. The 'undamaged' fuses might be 'half-blown' and will blow with the next high current.
- In case of serial fuses (e.g., DC fuses): If one of the serial fuses is blown, all serial fuses have to be changed. The 'undamaged' fuses might be 'half-blown' and will blow with the next high current.

DC-current pulses measured using an oscilloscope

Connect an oscilloscope at the fixed AO named IACT (XAO:4/5 on the SDCS-CON-K02L) and check for the proper number of current pulses:



There should be six current pulses in positive direction.

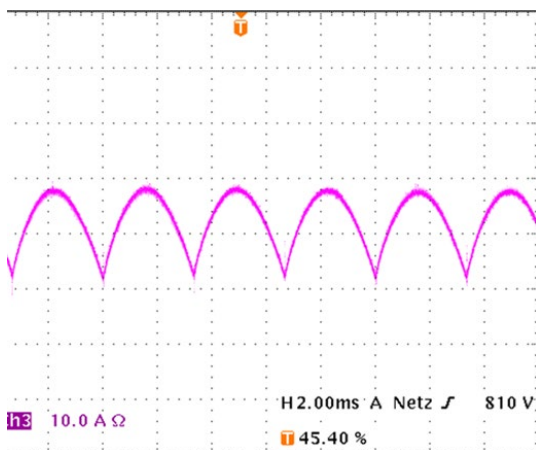
In case of a 4-Q drive the six current pulses for the negative current direction or other speed direction have to be checked.

Checking current bubbles using 30.44 Minimum firing angle

Drive Composer information:



- Connect an oscilloscope at the fixed AO named IACT (XAO:4/5 on the SDCS-CON-K02L).
- Connect Drive Composer to the drive and choose local mode.
- Monitor 27.02 Used current reference, 27.05 Motor current and 27.18 Firing angle.
- Set 27.22 Current reference source = 27.23 Current reference external.
- Set 99.07 M1 used field exciter type = None.
- Set 30.44 Minimum firing angle = 120°.
- Start the drive via Drive Composer.
- Set the armature current reference in 27.23 Current reference external to:
 - +50 % for positive current.
 - -50 % for negative current.
- Decrease the minimum firing angle degree by degree. At around 105° the current should start to flow. Now check the number of current bubbles. During a time of 20 ms (50 Hz mains)/16.66 ms (60 Hz mains) 6 bubble should be visible:



- Make sure the motor is not turning (**Attention:** let the drive run only for a brief time).

- Stop the drive via Drive Composer.
- Set 30.44 Minimum firing angle, 27.22 Current reference source and 99.07 M1 used field exciter type back to their original settings.

Thyristor test

Also, the thyristor test provided by the firmware can be used:

- Switch the drive to local mode (Drive Composer, control panel or local I/O).
- Start the thyristor test by means of 95.24 Service mode = Thyristor test and within 20 s set On and Run.
- During the thyristor test the mains contactor will be closed and the thyristors are checked. The field current is not released while the thyristor test is active and thus the motor should not turn.
- When the thyristor test is finished see warning AF90 for details.
- For more information consult the [DCS580 Firmware Manual \(3ADW000820\)](#).

Ripple monitor

The ripple monitor indicates that the ripple of the DC current is much higher than normal. In such a case, most often one thyristor does not work. Its missing current contribution causes a deep dip in the direct current.

The structure of the current loop (current controller) will force the other thyristors to compensate the dip by a certain overcurrent in order to keep the average current constant. Such compensation results in a ripple monitoring fault during motoring mode operation with $\alpha \leq 90^\circ$.

The reason for a thyristor without current may be:

- A blown line fuse.
- A loose gate/cathode connector.
- A thyristor does not get firing pulses or does not react to firing pulses.
- The current controller may be totally mismatched to the DC load.
- The AC mains network is causing that fault message. In this case, asymmetrical phase shift, uneven phase voltage or critical designed power factor correction equipment or harmonic reduction equipment can be the reason.

How to find a faulty thyristor

If a blown fuse is suspected, the problem is caused most often by a faulty thyristor. To make sure, that a thyristor is the reason and needs to be exchanged fault tracing must be done.



In general, make sure that all safety instructions given within this manual or within the Safety instructions, related to the machine or the application itself, are obeyed.

Drives size K1 ... K4 (20 ... 1000 A)

These drives require semiconductor fuses in the 3 AC lines.

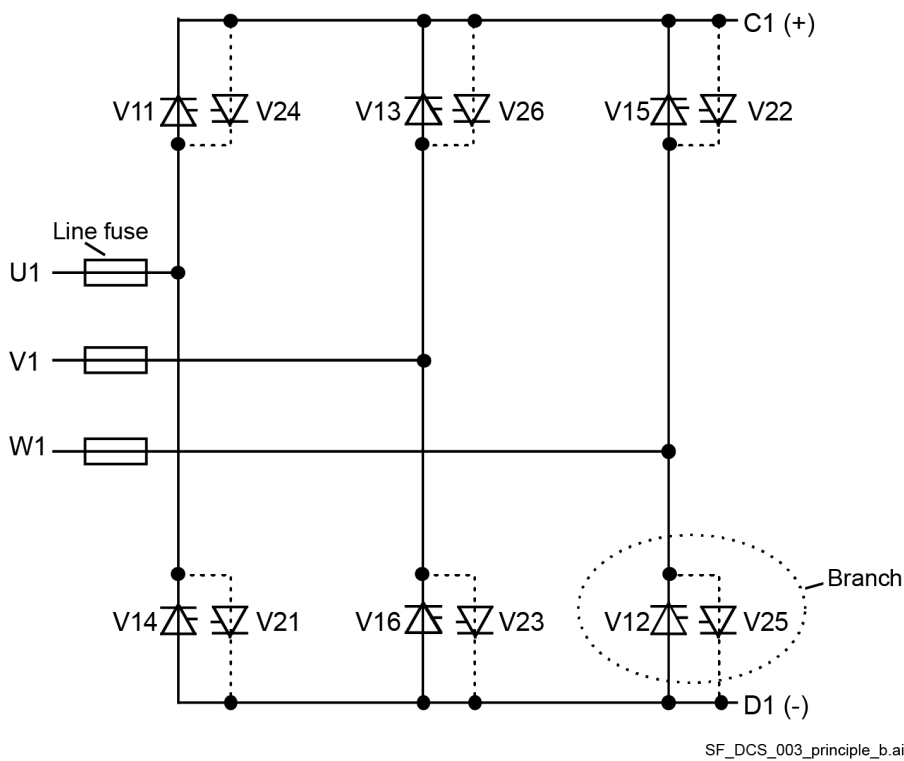
- The drive must be disconnected from the mains.
- One motor armature cable should be disconnected from the drive.

Blown fuses

- Make sure that the connection to the mains is open (e.g., open the mains breaker).
- Using the OHM function of a normal multimeter, measurements must be made over the line fuses.
- Normally, each measurement should show low resistance ($< 1 \text{ Ohm}$).

Blown thyristors

- Make sure that the connection to the motor is open (e.g., disconnect motor cables).
- Using the OHM function of a normal multimeter, measurements must be made from each AC terminal to each DC terminal (U1 to C1, V1 to C1, W1 to C1, U1 to D1, V1 to D1 and W1 to D1:



Bridge configuration K1 ... K4

- Normally, each measurement should show high resistance ($> 1 \text{ k}\Omega$).
- Target: Find a short circuit, indicated by low resistance ($< 1 \text{ }\Omega$) (destroyed thyristor).
- If the drive is designed with thyristor modules, then a module consists of two thyristors. In this case it is sufficient to know which thyristor module has a defective thyristor because the complete module must be replaced.
- After a thyristor module is replaced, the above-mentioned measurement should be done another time to make sure that all faulty thyristors have been detected!

Note: The RC/snubber circuit could also cause 0 Ω results for a short time.

The measurement, showing less resistance than 1 Ω , should be made a second time with test leads applied to the terminals with opposite polarity; if this measurement shows the same result, one or two thyristors located in that path are faulty; they need to be replaced.

Ripple monitor

If the ripple monitor fault occurred, a fault tracing as described above must be conducted:

- Check the fuses and the thyristors, according to the statements before.
- If the power section seems to be ok, but still one or more thyristors do not take current, something went wrong in between the firing pulse generation and the thyristor's gate; in this case check:
 - Is a firing pulse present on the primary side of the firing pulse transformer?
 - Is a firing pulse present on the secondary side of the firing pulse transformer?
 - Is the firing pulse transferred to the gate of the thyristor (loose gate connector)? Are all electrical connections still healthy?
 - Can the thyristor be fired with the applied firing pulse? Is the pulse form of the firing pulse identical at all measuring positions?
- Check the settings of the current controller.
- Check the AC mains network by taking recordings of the line voltage and current at all 3 phases at the same time.

How to detect a faulty SDCS-BAB-F0x (K1 ... K4)

SDCS-BAB-F0x problems can be noticed differently:

- Fault messages related to the excitation.
- The field current is oscillating.
- A fuse F100 ... F102 on the SDCS-PIN-K02 is blown (sizes K1 ... K3) or a fuse F401 ... F403 in the drive is blown (size K4). This is an indication that a strong overcurrent has happened due an internal short circuit in the bridge connected to the SDCS-BAB-F0x because of a defective thyristor or diode.

How to find a faulty SDCS-BAB-F0x

If a blown fuse on the SDCS-PIN-K02 or in the drive is suspected, the problem is caused most often by a faulty SDCS-BAB-F0x. To make sure that the SDCS-BAB-F0x is the reason and needs to be exchanged fault tracing must be done in two different ways, depending on the size of the drive.



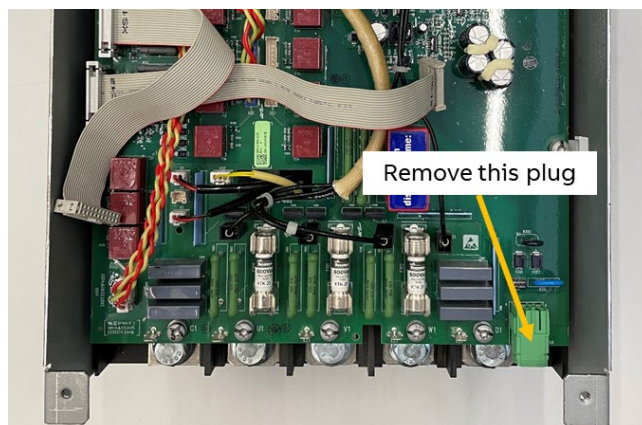
In general, make sure that all safety instructions given within this manual or within the Safety instructions, related to the machine or the application itself, are obeyed.

Drives size K1 ... K3 (20 ... 520 A)

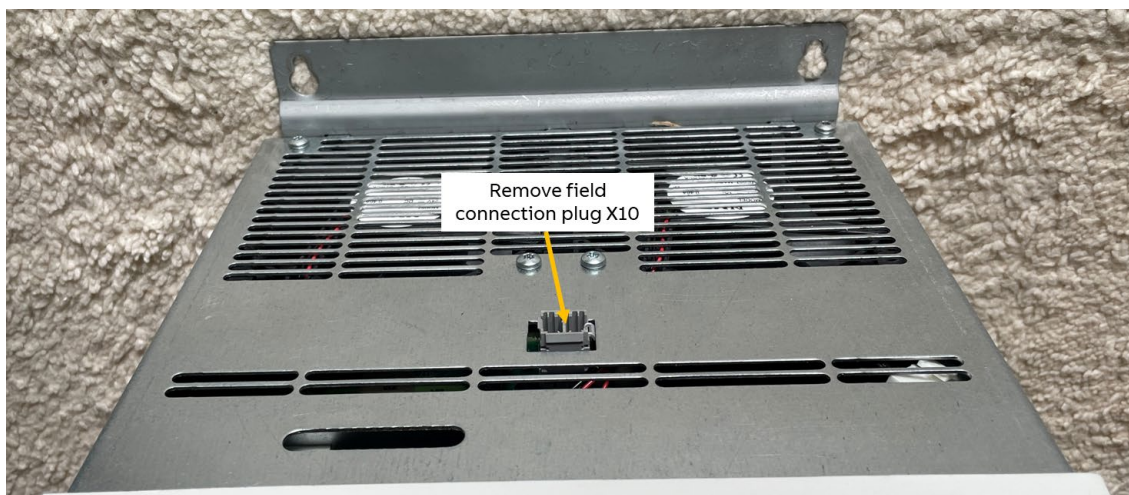
The supply for the SDCS-BAB-F0x is passed via the SDCS-PIN-K02. Thus, fuses F100 ... F102 need to be measured, before the SDCS-BAB-F0x is checked.

- The drive must be disconnected from the mains.
- Follow the instructions [Remove faulty thyristor modules](#) until step 8 is done.

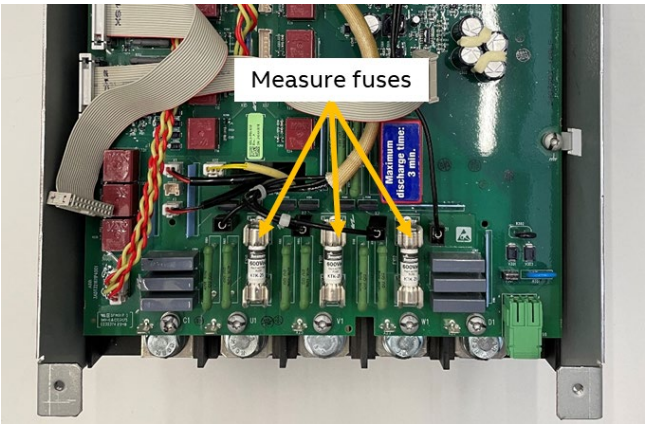
9. Remove the marked plugs at the SDCS-PIN-K02.



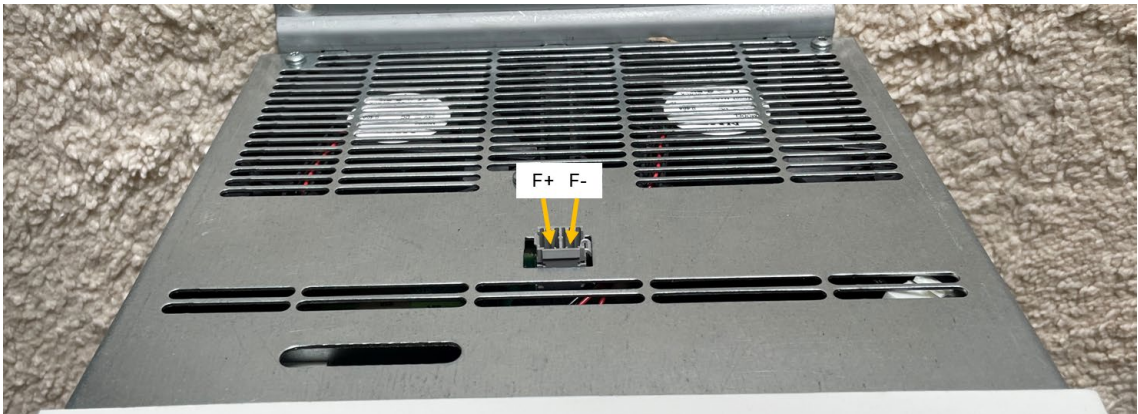
10. Remove the field connection plug X10 from the top of the DCS580.



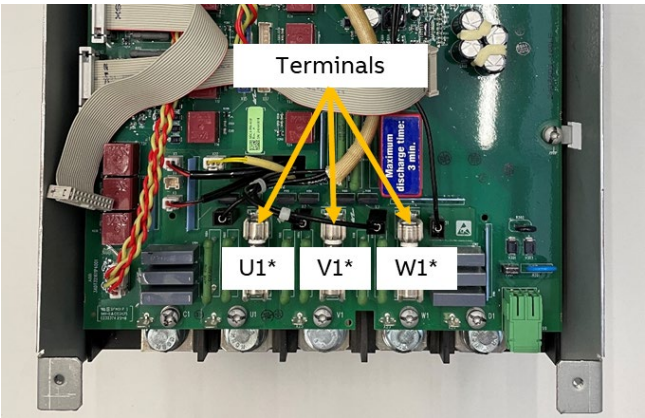
11. Measure fuses F100 ... F102 using the OHM function of a normal multimeter and exchange the broken ones.



12. The SDCS-BAB-F0x contains a 3-phase half-controlled bridge (B6H). Using the OHM function of a normal multimeter, measurements must be made from each AC terminal to each field exciter terminal (U1* - F+, V1* - F+, W1* - F+, U1* - F-, V1* F- and W1* -F-).



- Normally, each measurement should show high resistance ($> 1\text{ k}\Omega$).
- A short circuit is indicated by a low resistance ($< 1\text{ k}\Omega$, destroyed thyristor or diode).
- If one of the measurements shows a short circuit, replace the SDCS-BAB-F0x.
- A measurement, showing less resistance than $1\text{ }\Omega$ should be made for a second time. Apply the test leads with opposite polarity; if this measurement shows the same result, the SDCS-BAB-F0x is faulty and needs to be replaced.



- After the SDCS-BAB-F0x is replaced, the above-mentioned measurement should be made another time to make sure that all is fine.

	Fuses (F100 ... F102)	SDCS-BAB-F0x	To do
Faulty?	Yes	No	Exchange SDCS-PIN-K02
	No	Yes	Exchange SDCS-BAB-F0x
	Yes	Yes	Exchange SDCS-BAB-F0x and fuses

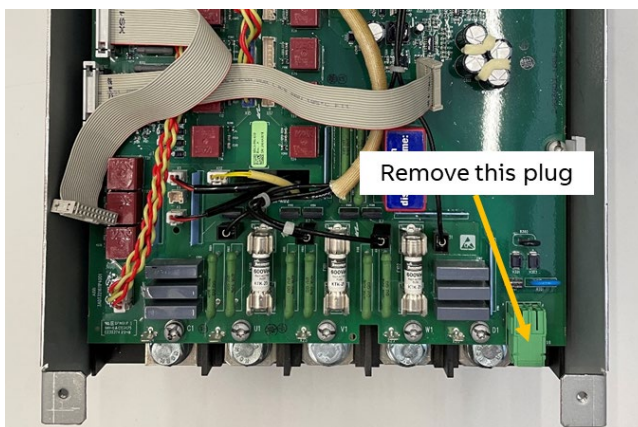
Exchange the SDCS-BAB-F0x continue with chapter [Remove faulty SDCS-BAB-F0x](#).
Exchange the SDCS-PIN-K02 continue with step 9 of the instructions [Remove faulty thyristor modules](#).

Drives size K4 (610 ... 1000 A)

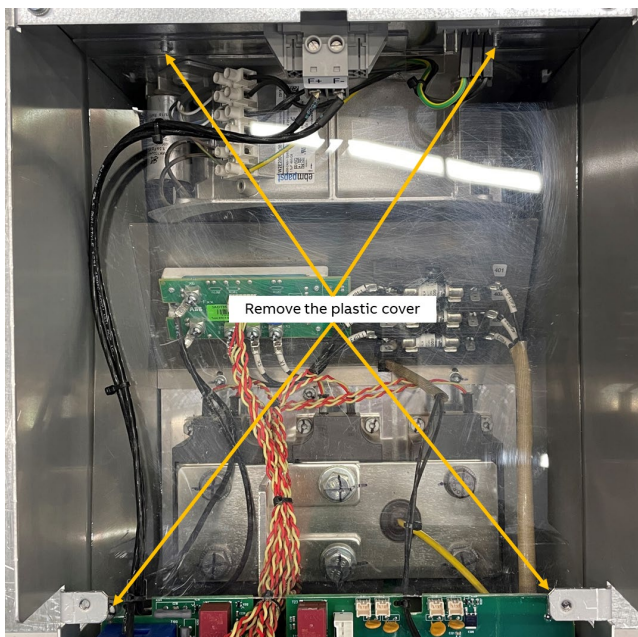
The supply for the SDCS-BAB-F0x is passed via fuses F401 ... F403. Thus, fuses F401 ... F403 need to be measured, before the SDCS-BAB-F0x is checked.

- The drive must be disconnected from the mains.
- Follow the instructions [Remove faulty thyristor modules](#) until step 8 is done.

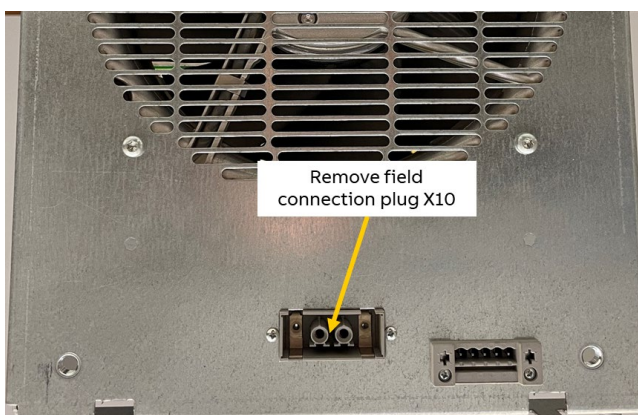
9. Remove the marked plugs at the SDCS-PIN-F02.



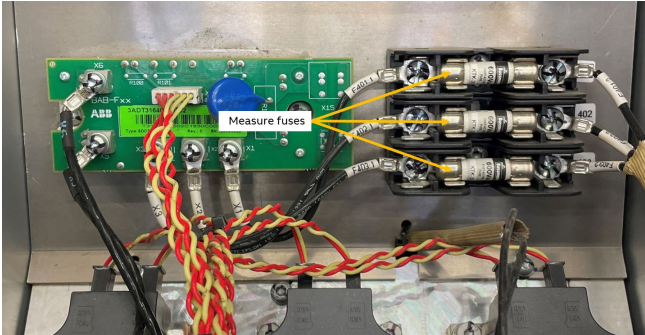
10. Remove the plastic cover.



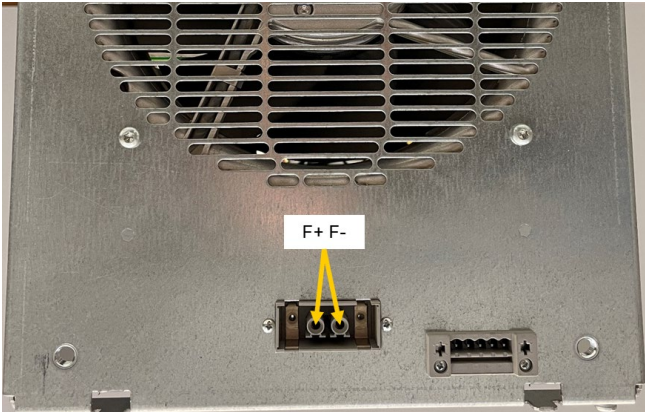
11. Remove the field connection plug X10 from the top of the DCS580.



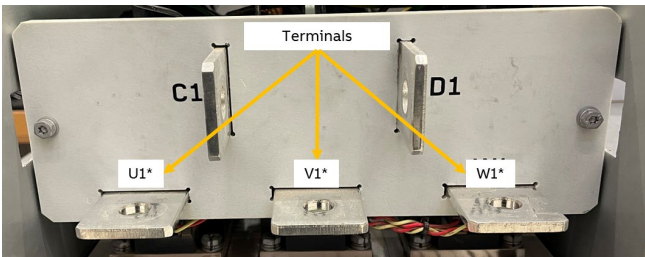
12. Measure fuses F401 ... F403 using the OHM function of a normal multimeter and exchange the broken ones.



13. The SDCS-BAB-F0x contains a 3-phase half-controlled bridge (B6H). Using the OHM function of a normal multimeter, measurements must be made from each AC terminal to each field exciter terminal (U1* to F+, V1* to F+, W1* to F+, U1* to F-, V1* to F- and W1* to F-).



- Normally, each measurement should show high resistance ($> 1\text{ k}\Omega$).
- A short circuit is indicated by a low resistance ($< 1\text{ k}\Omega$, destroyed thyristor or diode).
- If one of the measurements shows a short circuit, replace the SDCS-BAB-F0x.



- A measurement, showing less resistance than $1\text{ }\Omega$ should be made for a second time. Apply the test leads with opposite polarity; if this measurement shows the same result, the SDCS-BAB-F0x is faulty and needs to be replaced.
- After the SDCS-BAB-F0x is replaced, the above-mentioned measurement should be made another time to make sure that all is fine.

	Fuses (F401 ... F403)	SDCS-BAB-F0x	To do
Faulty?	Yes	No	Exchange fuses
	No	Yes	Exchange SDCS-BAB-F0x
	Yes	Yes	Exchange SDCS-BAB-F0x and fuses

Exchange the SDCS-BAB-F0x continue with chapter [Remove faulty SDCS-BAB-F0x](#).

Handling the semiconductors

General Instruction how to handle semiconductors

Thyristor modules in drives size K1 ... K4 (20 ... 1000 A), busbars and fuses have to be mounted with the correct torque using a torque screwdriver or a torque wrench.

Always mark suspected damaged components clearly after removing them from the circuit, to avoid confusion with "good" components.

When removing a damaged semiconductor, write down how and where it was installed (direction, location and connected gate leads).

Check that the new and old components have the same type designation or that the new component can replace the old one. A semiconductor can be replaced by different compatible semiconductors according to the codes in the manufacturers' table.

Semiconductor components are high-precision products. All unnecessary used tools and objects might damage the easily dented and scratched surfaces of the semiconductors.

1. Keep new semiconductors as long as possible in their original packages.
2. Use protective gloves if possible.
3. Clean work area and hands frequently.
4. Use good illumination.

Exchange thyristors sizes K1 ... K4

Installation of thyristor modules and SDCS-BAB-F0x in drives size K1 ... K4 (20 ... 1000 A)

All DCS580 size K1 ... K4 are equipped with thyristor modules and a SDCS-BAB-F0x for excitation. In order to keep the operating temperature of the semiconductor module low, the joint between the heat sink and the module should have good heat conducting ability. The electrical conductivity of the connectors must also be good. For this reason, the following instructions must be observed with particular care.

Required tools

Special tools or material needed in addition to standard tools for the exchange of thyristor modules:

- Torx screwdrivers: TX10, TX20, TX25
- Torque spanner: Mounting torques for the thyristor modules and SDCS-BAB-F0x to heat sink and electrical connections see table [Nominal mounting torque for thyristor modules and SDCS-BAB-F0x](#).
- Torque spanner for electrical connections: 13Nm (M8)
25 Nm (M10)
50 Nm (M12)
- Screws are metric type; use appropriate nuts.
- Tissue paper.
- Solvent (e.g., ethanol).
- Thermal joint compound (non-conducting) for thyristor modules: GHSN390011P0010
Type Berulub FZ1 E3
or
- Thermal joint compound (conducting) for thyristor modules and disk type thyristors: GHSN390001P0001
Type Berulub FK-SU 2



Before the work starts, disconnect the drive completely from the power supply, then check the voltage free condition and make sure everything is located in an electrical and mechanical safe condition!

Find faulty thyristor modules

See [Fault Tracing Thyristors](#) of this publication.

Remove faulty thyristor modules



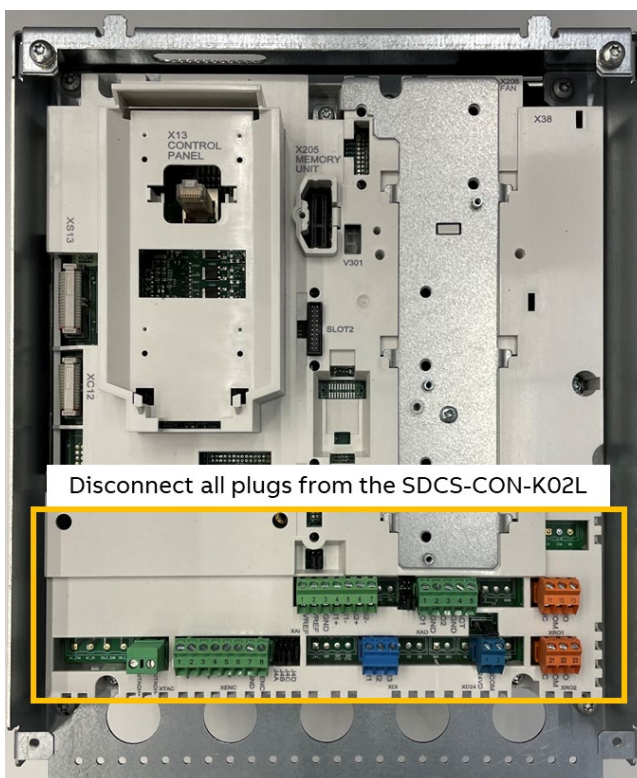
Recommendation:

Use safety equipment to protect your hands.

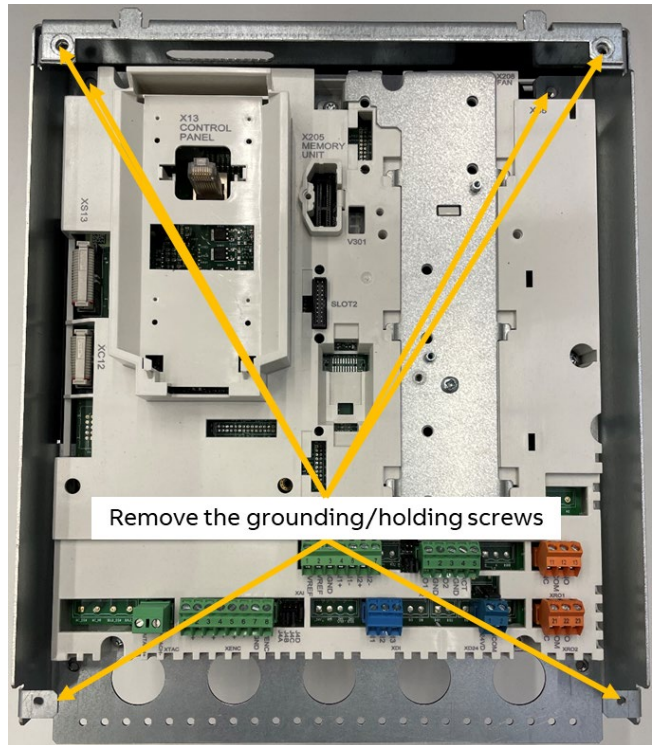
1. Remove control panel and design cover.



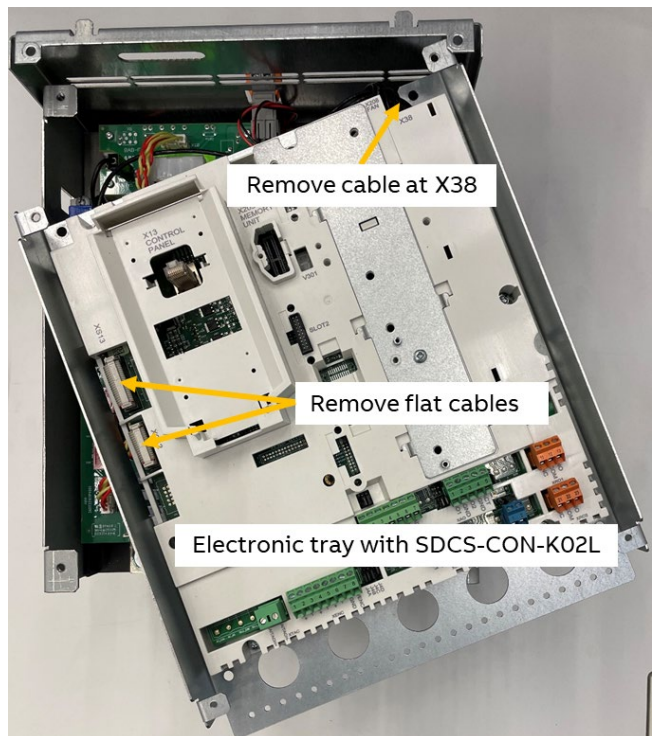
2. Disconnect all I/O plugs from the electronic tray.
3. Disconnect all connections from present plug-in options.



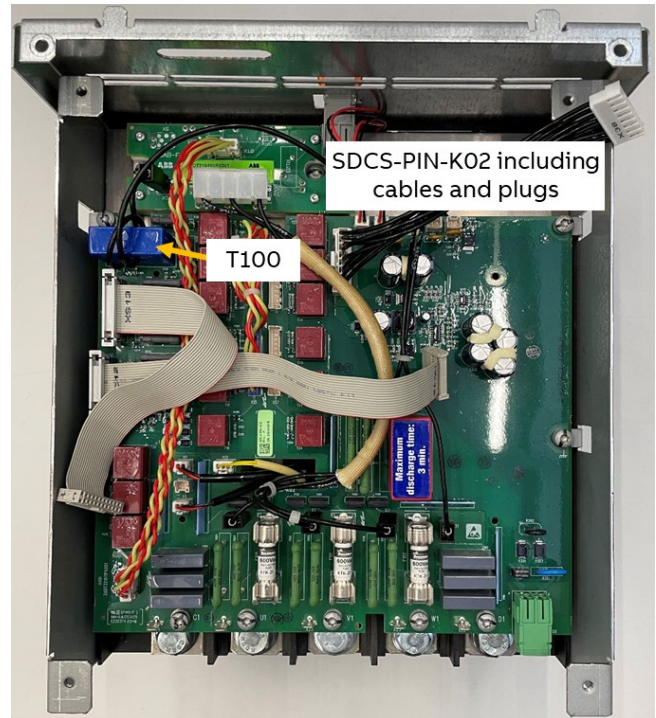
4. Remove the grounding/holding screws from the electronic tray.



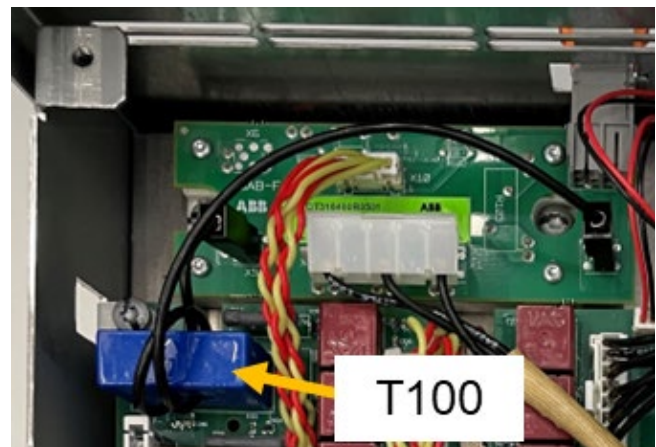
5. Remove the electronic tray.
6. Before removing the electronic tray completely unplug the flat cables (XC12, XS13) and the cable connected at X38.
7. Remove the electronic tray completely.



8. The result looks like this.
9. Remove all cables and plugs at the SDCS-PIN-K02.
 - X99 auxiliary power supply.
 - XC12, XS13, X38 flat cables/cable from SDCS-CON-K02L to SDCS-PIN-K02.
 - X4, X5/X6, X15/X10 on the SDCS-BAB-F0x.
 - X30, X31 snubber resistor.
 - X41 ... X44 fans supply.
 - X1, X2, X7 line voltage for SDCS-BAB-F0x.
 - X11 firing pulses for SDCS-BAB-F0x.
 - X15 ... X18 gate leads.
 - X3 ... X5 current transformers.
 - X22 temperature sensor.



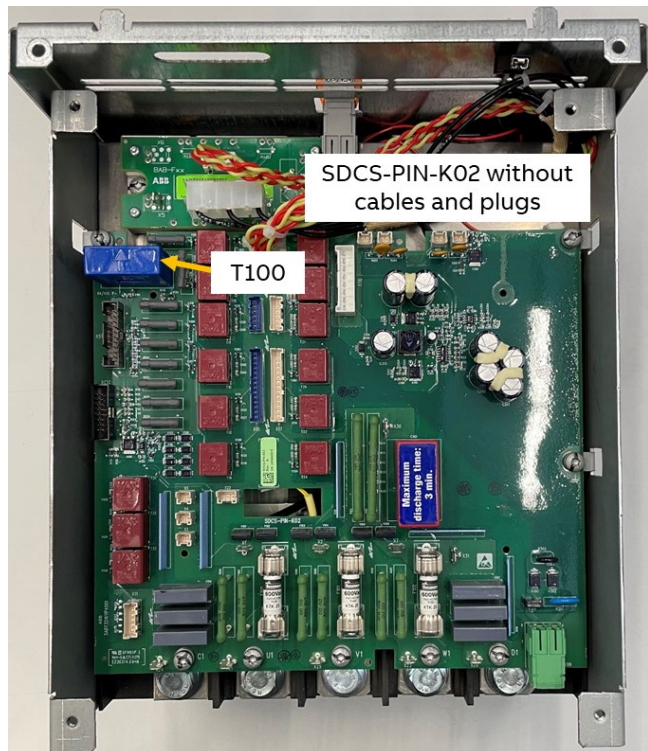
10. Keep the winding direction and the number of windings through T100 in mind:



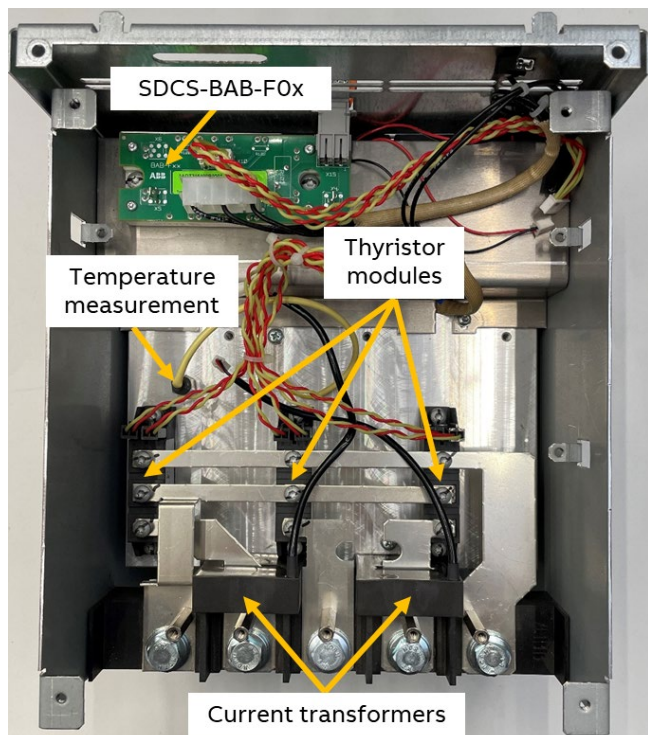
Size	Drive type	Used type	Used fuses	T100 threads	IF [A]
K1	DCS580-S01-0020 ... DCS580-S02-0100	SDCS-BAB-F01	F100 ... F102 on SDCS-PIN-K02 KTK 25 = 25 A	3①	1 ... 12
K2	DCS580-S01-0135 ... DCS580-S02-0300	SDCS-BAB-F01	F100 ... F102 on SDCS-PIN-K02 KTK 25 = 25 A	2①	1 ... 18
K3	DCS580-S01-0315 ... DCS580-S02-0520	SDCS-BAB-F02	F100 ... F102 on SDCS-PIN-K02 KTK 25 = 25 A	1①	2 ... 25
K4	DCS580-S01-0610 ... DCS580-S02-1000	SDCS-BAB-F02	F401 ... F403 in drive KTK 30 = 30 A	1①	2 ... 30

① Number of threads through the hole in the T100 (e.g., 3 threads equal 2 loops).

11. The result should look like this.



12. Remove the SDCS-PIN-K02.



13. Remove the gate leads from the faulty thyristor module and mark the connectors clearly.
14. Remove the busbars necessary to get full access to the faulty thyristor module.
15. If a current transformer must be removed, mark its position, direction, and the connections clearly.
- Note:** Remove only as many parts as needed around the faulty thyristor module.
16. Remove the faulty thyristor module and mark it clearly as defective.

Exchange thyristors sizes K1 ... K4

Install new thyristor modules

1. Ensure that the new thyristor module is of the correct type.
2. Remove old heat-conducting compound (grease) from the heat sink. Clean the mounting surfaces (heat sink and thyristor module) with an appropriate solvent (e.g., ethanol) by means of tissue paper. When the heat sink is clean, spread out the heat-conducting compound with a rubber spatula or by hand.
3. Apply a **thin** layer of heat-conducting compound to the new thyristor module.



4. Spread the heat-conducting compound evenly by moving the thyristor module forward and backward on the heat sink.
5. Tighten all clamping screws by hand until the screw heads touch the bottom of the thyristor module. Then pre-torque the screws to 2.0 Nm.
Note: If the thyristor module is mounted by means of four screws, tighten the screws crosswise.
6. Tighten the screws to nominal torque according to table [Nominal mounting torque for thyristor modules and SDCS-AB-F0x](#).

Thyristor modules and SDCS-BAB-F0x		Nominal mounting torque	
Size (width)	Type	Electrical connections	Thyristor module to heat sink
20 mm block	SKKT27, 57, 106	2.6 ... 3.4 Nm	4.25 ... 5.75 Nm
20 mm block	MCC26, 56, 95	2.5 ... 4 Nm	2.5 ... 4 Nm
34 mm block	MCC162, 200	4.5 ... 5.5 Nm	2.25 ... 2.75 Nm
50 mm block	MCC255, 312	11 ... 13 Nm	4.5 ... 7 Nm
50 mm block	TT250	10.8 ... 13.2 Nm	4.25 ... 5.75 Nm
60 mm block	MCC501	10.2 ... 13.8 Nm	4.25 ... 5.75 Nm
60 mm block	MT3	10.8 ... 13.2 Nm	5.1 ... 6.9 Nm
60 mm block	TT425	10.8 ... 12.6 Nm	5.1 ... 6.9 Nm
-	SDCS-BAB-F01	-	2.7 ... 3.3 Nm
-	SDCS-BAB-F02	1.9 ... 2.1 Nm	2.7 ... 3.3 Nm

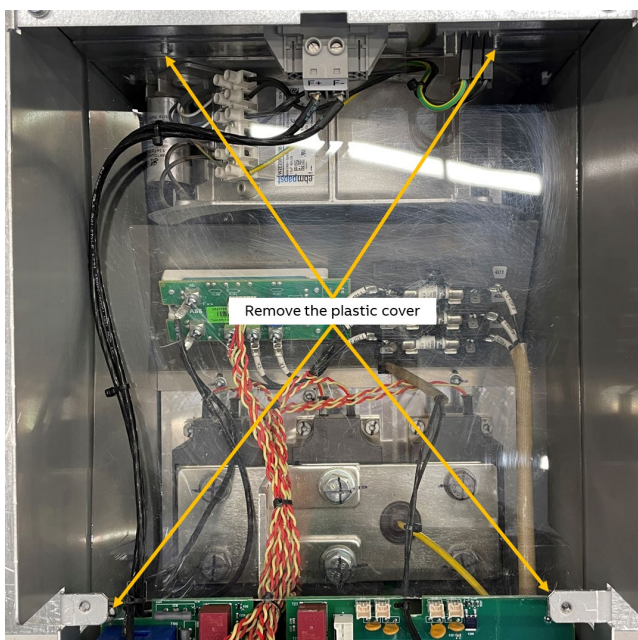
7. Reinstall the current transformer. Make sure its position and direction is correct.
8. Reinstall the busbars. Make sure, the correct torque is applied according to table [Nominal mounting torque for thyristor modules and SDCS-BAB-F0x](#).
9. Reconnect all gate leads to the thyristor module.
10. Perform an OHM test to make sure the thyristor is ok.
11. Reinstall the SDCS-PIN-K02 board.
12. Reconnect all cables and plugs at the SDCS-PIN-K02:
 - X22 temperature sensor.
 - X3 ... X5 current transformers.
 - X15 ... X18 gate leads.
 - X11 firing pulses for SDCS-BAB-F0x.

- X1, X2, X7 line voltage for SDCS-BAB-F0x.
 - X41 ... X44 fans supply.
 - X30, X31 snubber resistor.
 - X4, X5/X6, X15/X10 on the SDCS-BAB-F0x.
 - XC12, XS13, X38 flat cables/cable from SDCS-CON-K02L to SDCS-PIN-K02.
 - X99 auxiliary power supply.
13. At the SDCS-CON-K02L reconnect the flat cables (XC12, XS13), the cable connected at X38 and re-mount the electronic tray.
 14. Reconnect the grounding/holding screws at the electronic tray.
 15. Reconnect all I/O plugs with the electronic tray and all connections at the present plug-in options.
 16. Reinstall the design cover and the control panel.

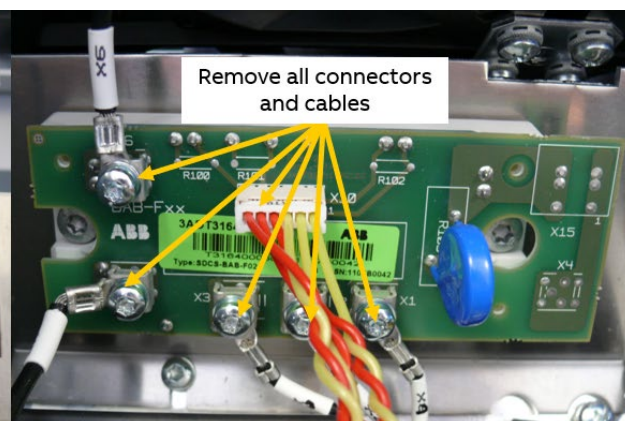
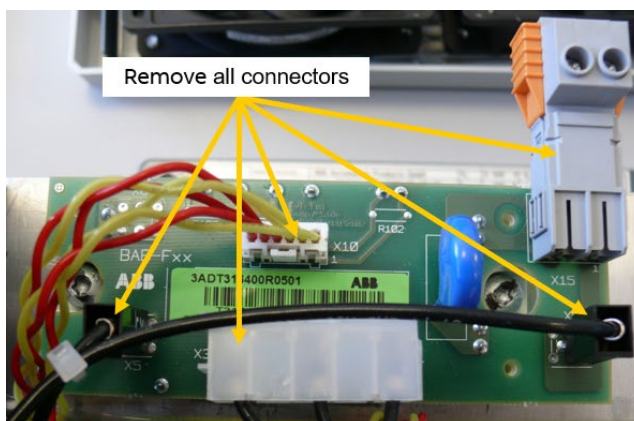
Remove faulty SDCS-BAB-F0x

- Follow the instructions Remove faulty thyristor modules until step 8 is done.

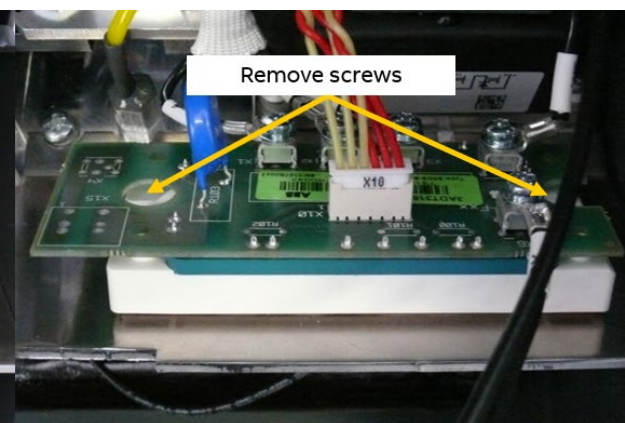
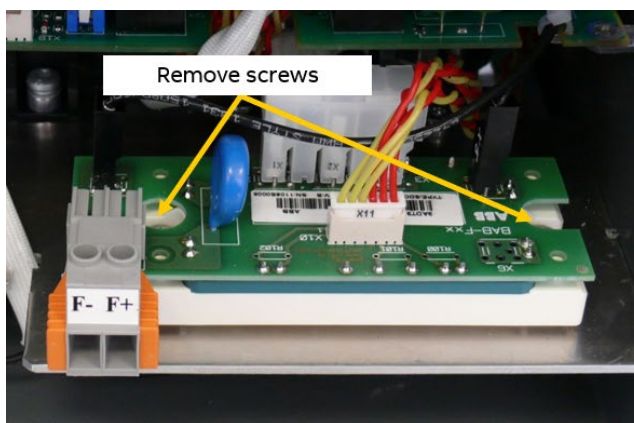
9. If it exists, remove the plastic cover.



10. Remove all connectors (SDCS-BAB-F01) or cables (SDCS-BAB-F02) from the faulty SDCS-BAB-F0x and mark them clearly.



11. Remove the faulty SDCS-BAB-F0x and mark it clearly as defective.



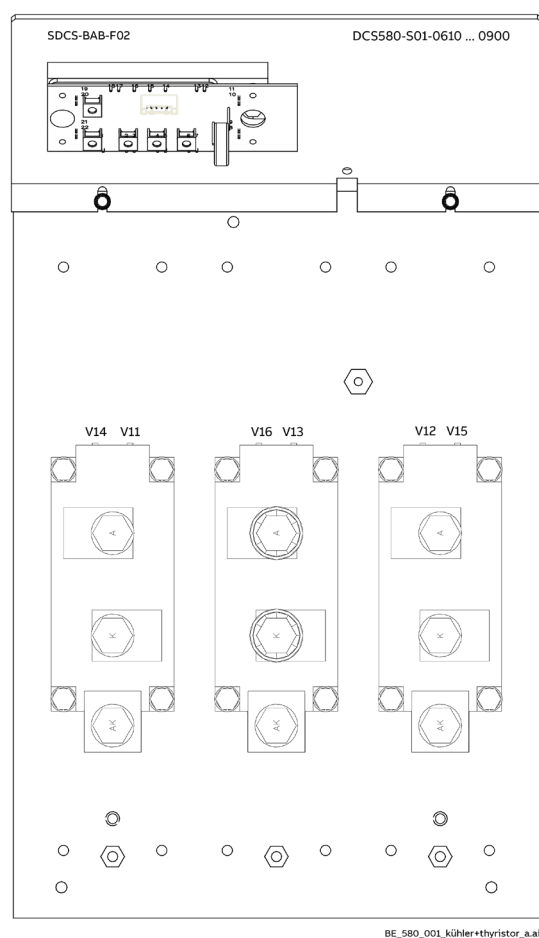
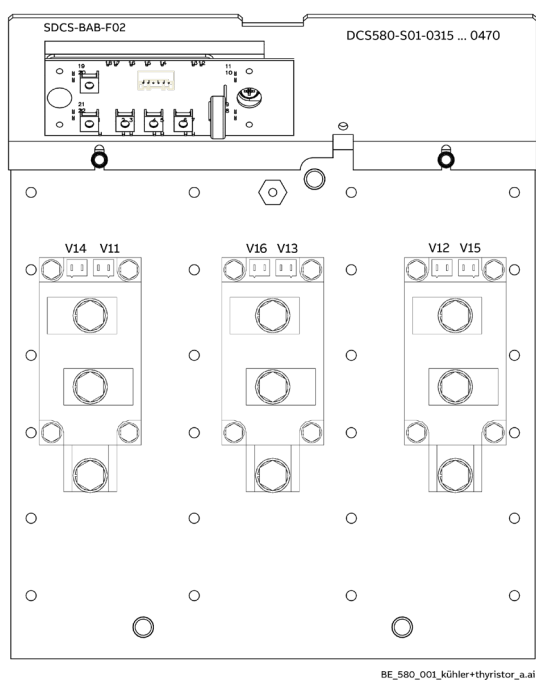
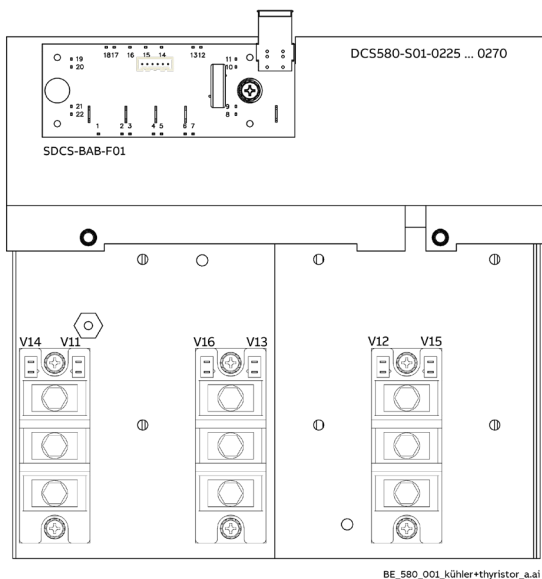
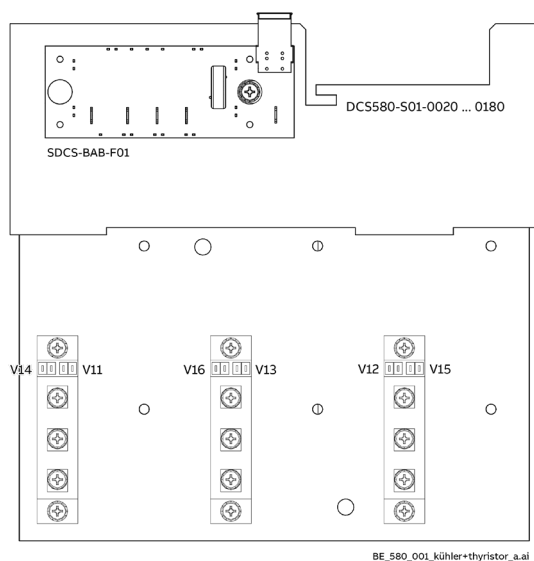
Exchange thyristors sizes K1 ... K4

Install new SDCS-BAB-F0x

1. Ensure that the new SDCS-BAB-F0x is of the correct type, either SDCS-BAB-F01 or SDCS-BAB-F02.
2. Remove old heat conducting compound (grease) from the heat sink. Clean the mounting surfaces (heat sink and SDCS-BAB-F0x bridge) with an appropriate solvent (e.g., ethanol) by means of tissue paper. When the heat sink is clean, spread out the heat conducting compound with a rubber spatula or by hand.
3. Apply a thin layer of heat conducting compound to the new SDCS-BAB-F0x bridge.
4. Spread the heat conducting compound evenly by moving the SDCS-BAB-F0x forward and backward on the heat sink.
5. Tighten all clamping screws by hand until the screw heads touch the bottom of the thyristor module. Then tighten the screws to 2.0 Nm torque.
6. Tighten the screws to nominal torque according to table [Nominal mounting torque for thyristor modules and SDCS-BAB-F0x](#).
7. Reconnect all connectors or cables to the SDCS-BAB-F0x.
8. If existing, reconnect the plastic cover.
9. Follow the instructions [Install new thyristor modules](#) beginning with step 13.

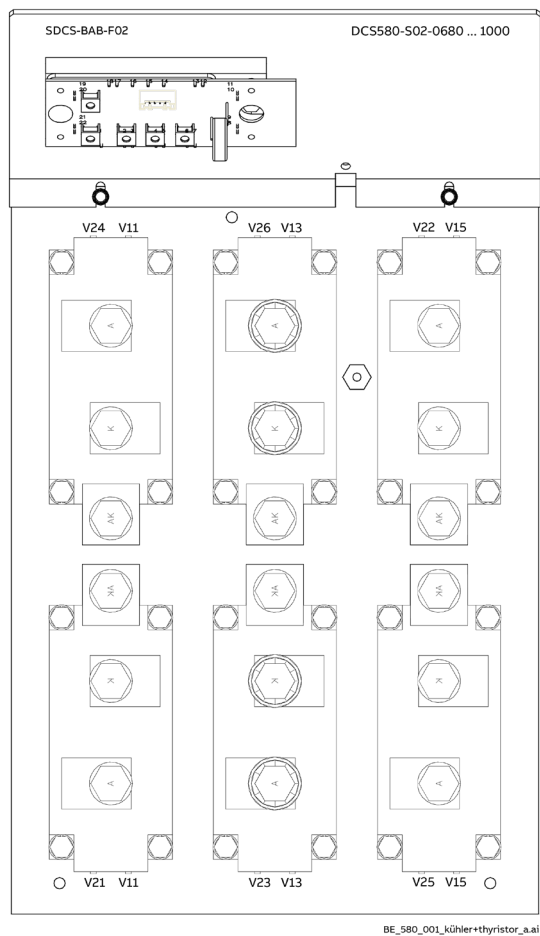
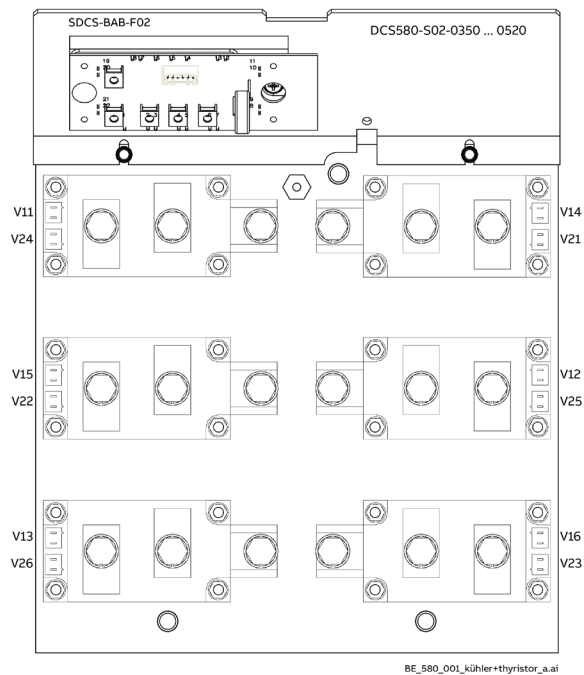
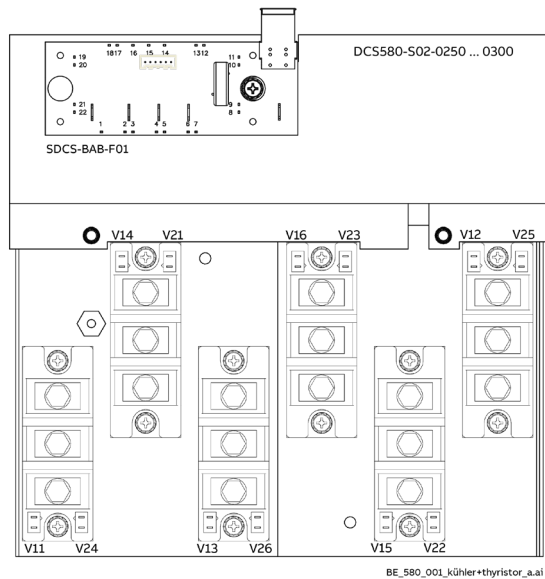
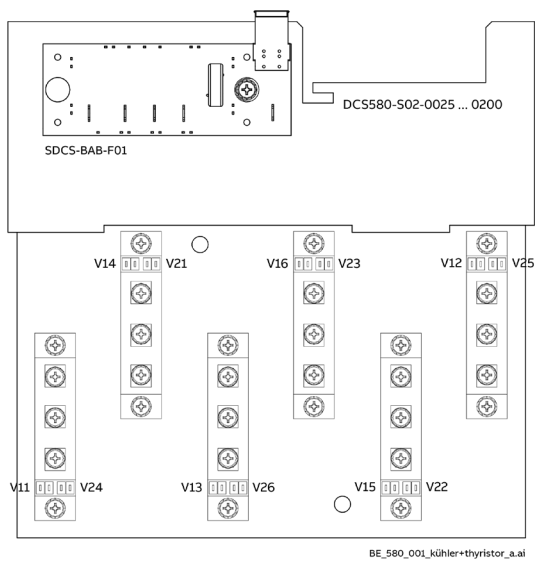
Thyristor module and SDCS-BAB-F0x location in DCS580-S01 (2-Q) drives

525 V_{AC} types:



Thyristor module and SDCS-BAB-F0x location in DCS580-S02 (4-Q) drives

525 V_{AC} types:



Exchange thyristors sizes K1 ... K4

Thyristor module terminals and SDCS-BAB-F0x

The terminal description is stamped or marked by a sticker on all thyristor modules and the bridge connected at the SDCS-BAB-F0x.

For all firing pulse cables is valid:

- **Yellow** is gate lead.
- **Red** is cathode lead.

Exchange the electronic tray (SDCS-CON-K02L)

General

All DCS580 size K1 ... K4 are equipped with an electronic tray DCS580 (includes SDCS-CON-K02L). Thus, the exchange is similar for all drive sizes.

Required Tools

Special tools or material needed in addition to standard tools for the exchange of the electronic tray:

- Torx screwdrivers: TX10, TX20, TX25.



Before the work starts, disconnect the drive completely from the power supply, then check the voltage free condition and make sure everything is located in an electrical and mechanical safe condition!

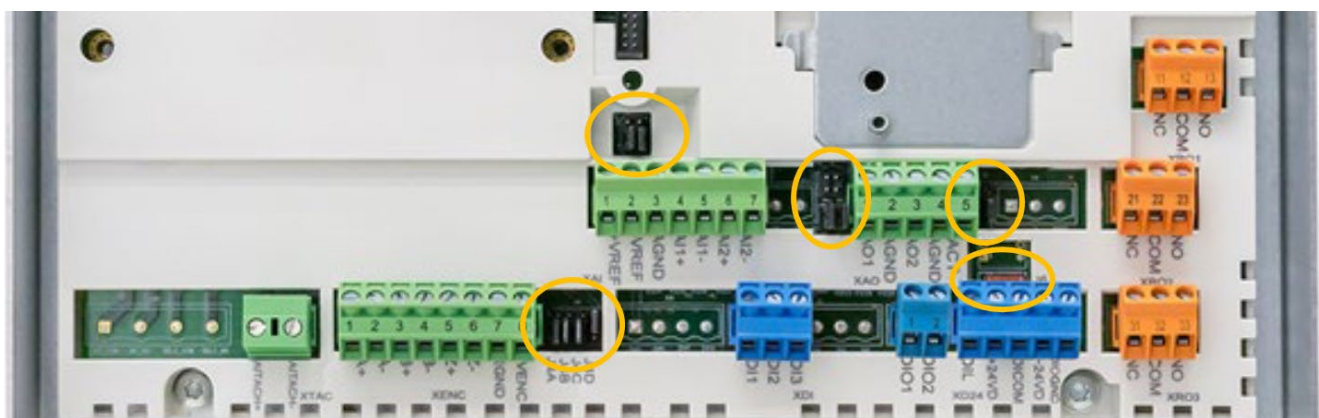
Overview of electronic tray exchange

1. Write down the type code of the drive.
2. Set the jumpers on the new electronic tray according to the old electronic tray.
3. Exchange the electronic tray.
4. Set type code of the drive.

1. Write down the type code of the drive

Write down the contents of 95.25 Set: Type code. This information must be set manually in the new electronic tray.

2. Set the jumpers on the new electronic tray according to the old electronic tray



Copy the jumper settings (J1 ... J7) of the old electronic tray onto the new electronic tray accordingly.

3. Exchange the electronic tray

Remove the old electronic tray

1. Remove control panel and design cover.



2. Remove the memory unit from X205.

The drive is equipped with a memory unit that is plugged into X205 on the electronic tray. The memory unit contains the firmware and the parameters.

When an electronic tray is replaced, the parameter settings can be retained by transferring the memory unit from the defective electronic tray to the new electronic tray.

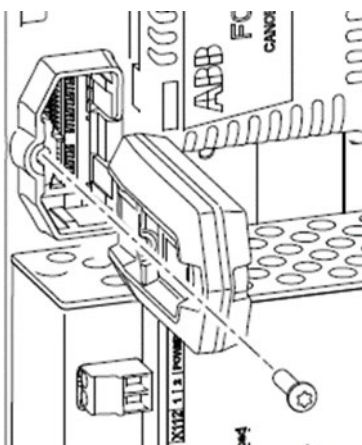


WARNING

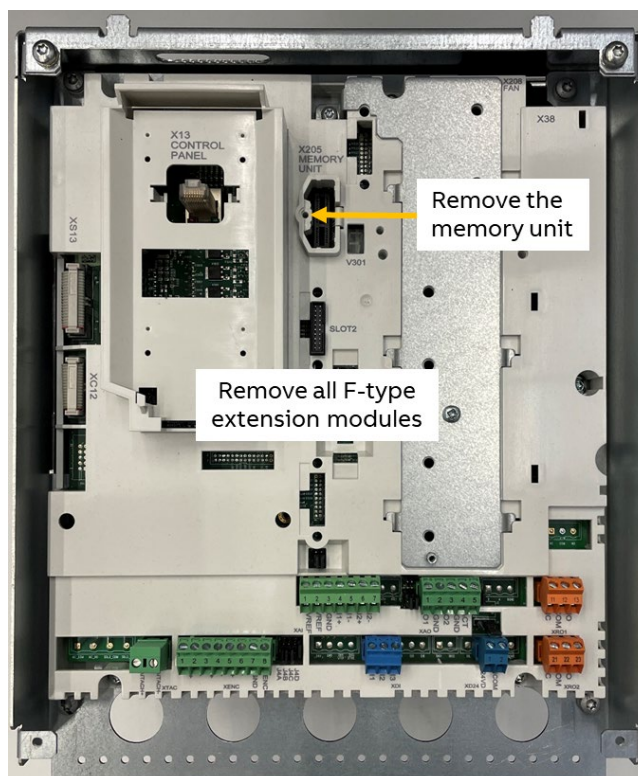
Do not remove or insert a memory unit when the drive is powered.

Replacing the memory unit

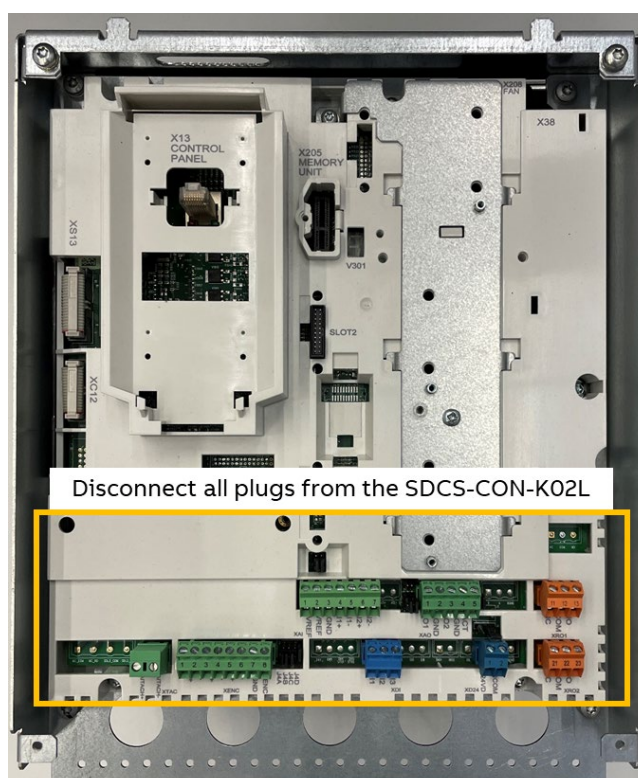
Unscrew the memory unit and pull it out.
Replace the memory unit in reverse order.



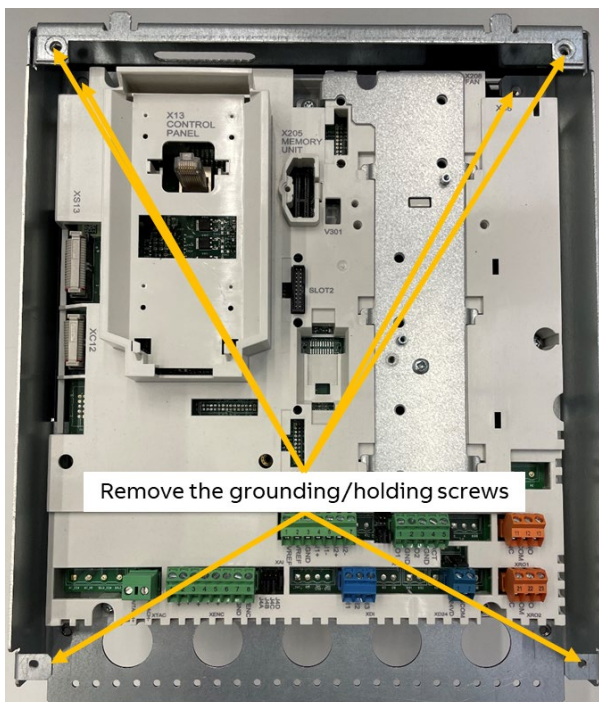
3. Remove all F-type extension modules.



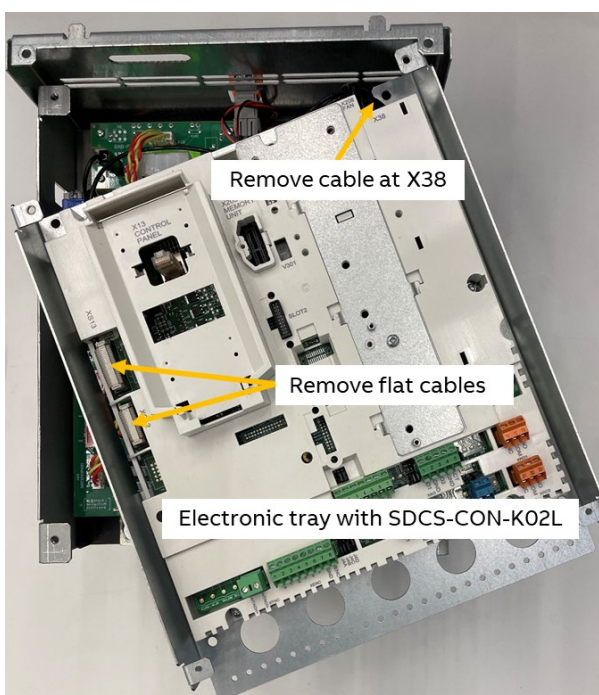
4. Disconnect all I/O plugs from the electronic tray.



5. Remove the grounding/holding screws from the electronic tray.



6. Remove the electronic tray.
7. Before removing the electronic tray completely unplug the flat cables (XC12, XS13) and the cable connected at X38.
8. Remove the electronic tray completely.



Install the new electronic tray

1. At the SDCS-CON-K02L reconnect the flat cables (XC12, XS13), the cable connected at X38 and re-mount the electronic tray.
2. Reconnect the grounding/holding screws at the electronic tray.
3. Reconnect all I/O plugs to the electronic tray.
4. Reinstall all plug-in options (do not forget the screws), the memory unit, the design cover, and the control panel.

4. Set type code of the drive

Follow the instructions in chapter [Set type code](#).

Service

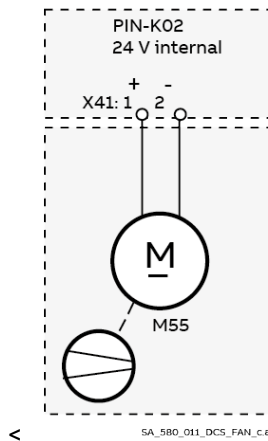
Exchange cooling fan size K1 (one fan)



Before the work starts, disconnect the drive completely from the power supply, then check the voltage free condition and make sure everything is located in an electrical and mechanical safe condition!

Drives size DCS580-S01-0020-05 and DCS580-S02-0025-05 have no cooling fans. They are convection cooled.

Drives size DCS580-S0b-0045-05 ... DCS580-S0b-0100-05 using one 24 V_{DC} fan.

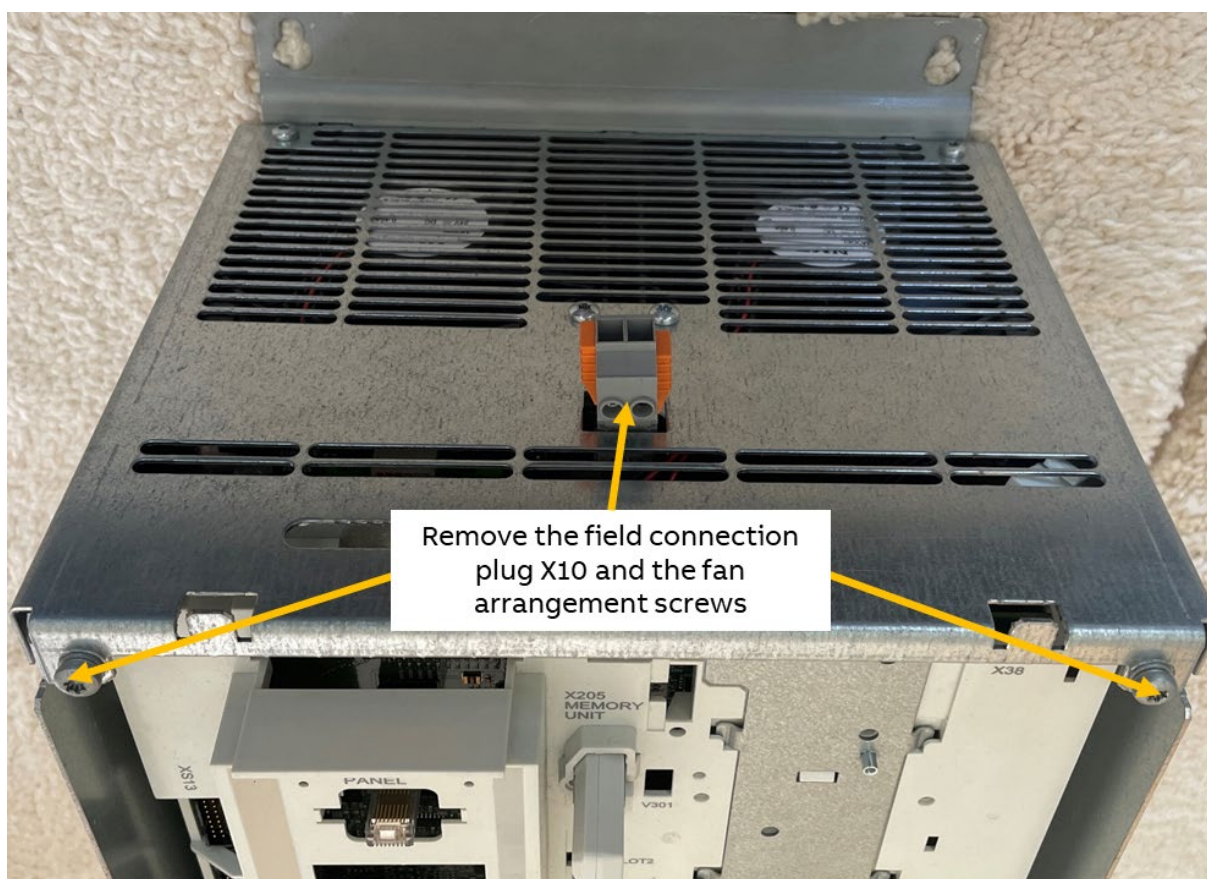


Removing the cooling fan

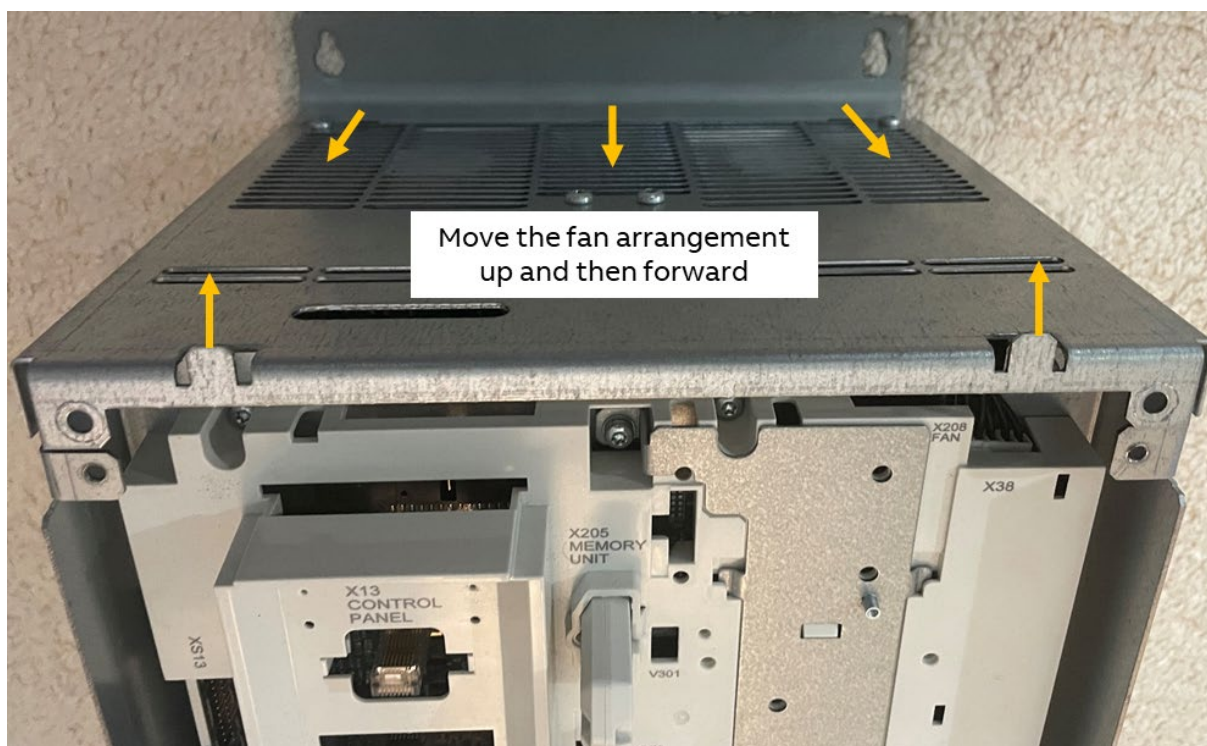
1. Remove the control panel and the design cover.



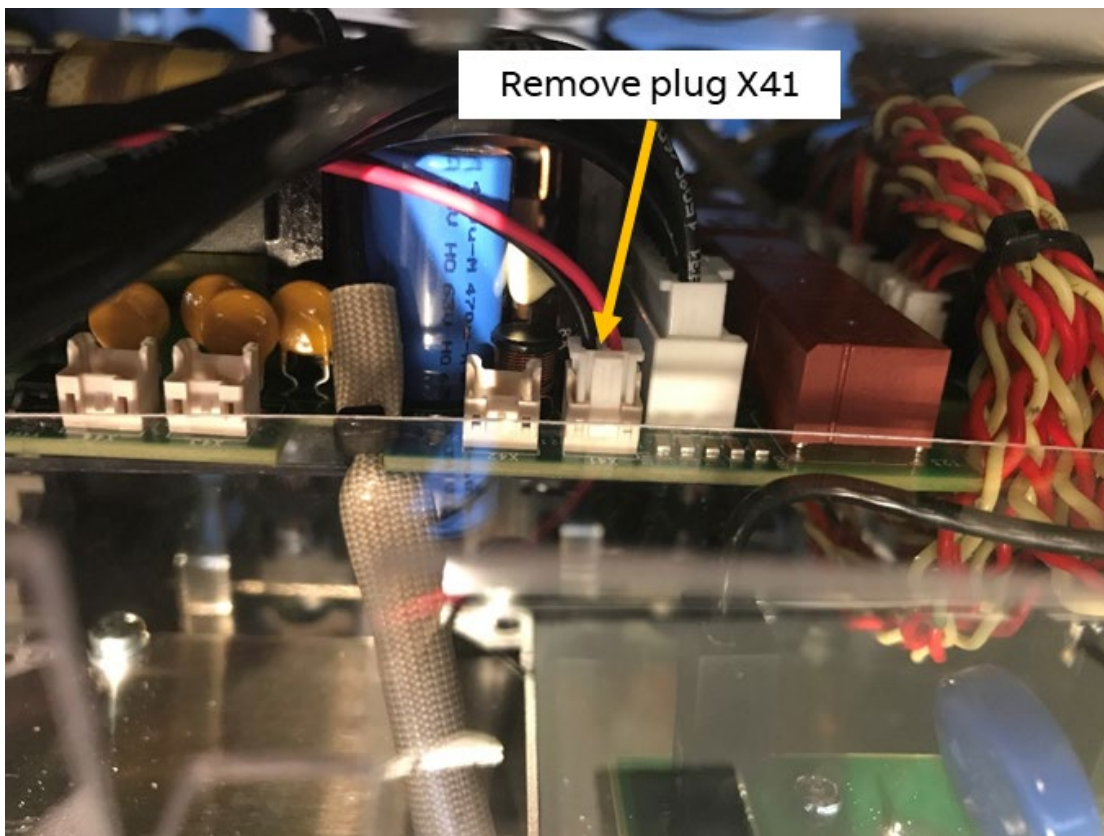
2. Remove the field connection plug X10 and the two fan arrangement screws.



3. Move the fan arrangement up and then forward.



4. Remove plug X41 on the SDCS-PIN-K02.



5. Remove the fan holding screws of the broken fan.



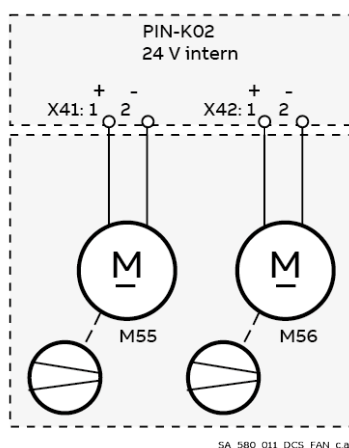
6. Attach the new fan, reassemble all parts, and check for correct blow direction of the fan. The air should be sucked out of the module.

Exchange cooling fans sizes K2 ... K3 (two fans)



Before the work starts, disconnect the drive completely from the power supply, then check the voltage free condition and make sure everything is located in an electrical and mechanical safe condition!

Drives size DCS580-S0b-0135-05 ... DCS580-S0b-0450-05 and using two 24 V_{DC} fans.

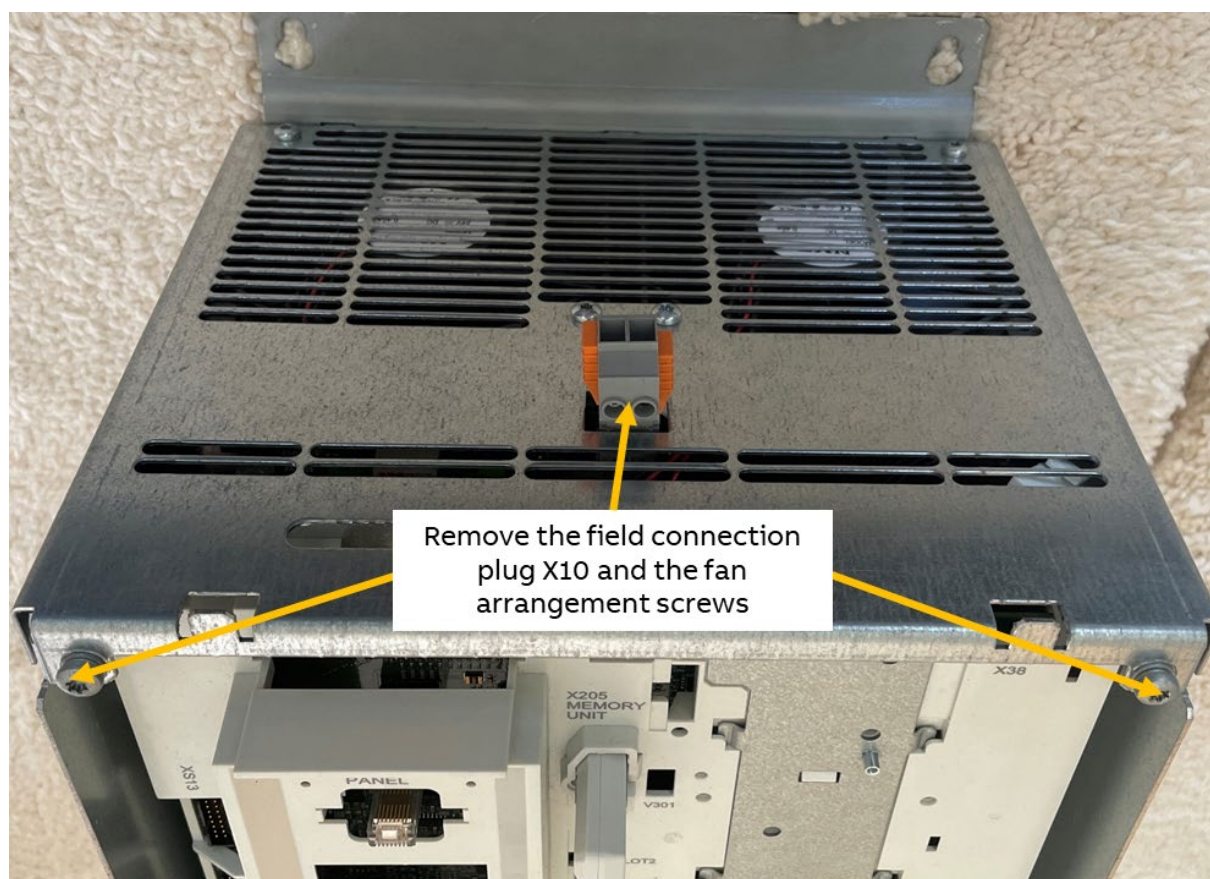


Removing the cooling fan

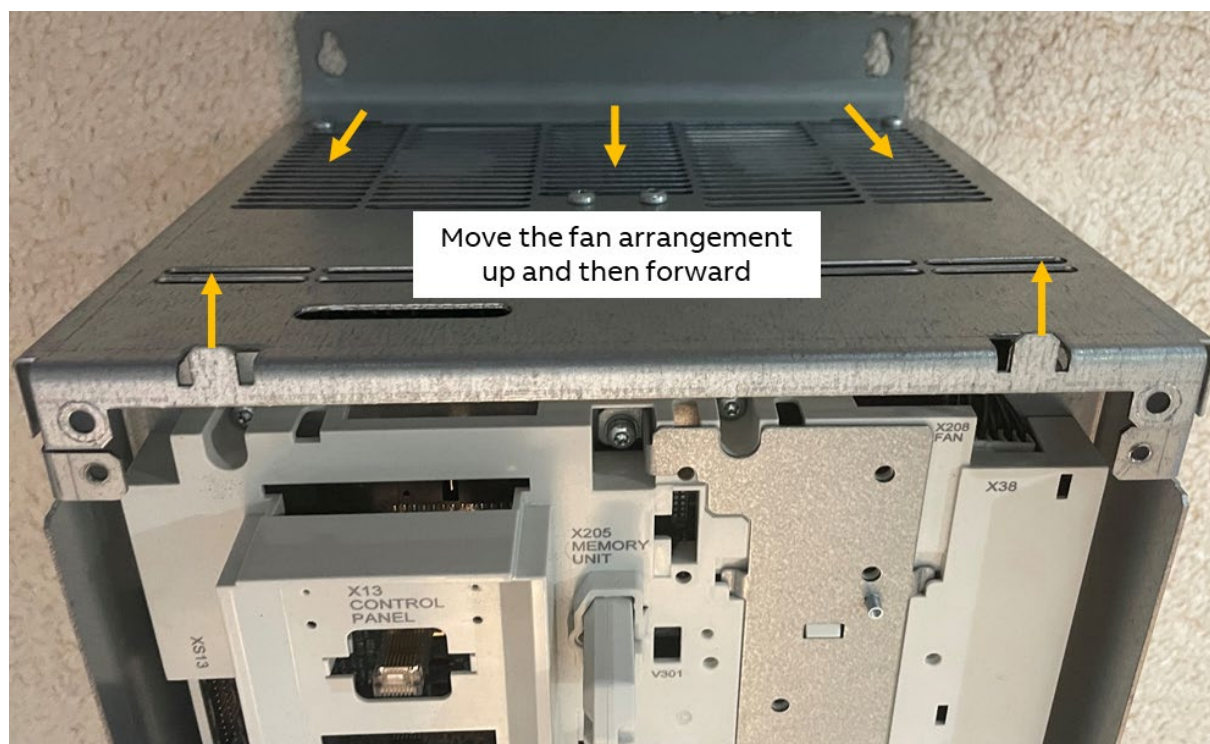
1. Remove the control panel and the design cover.



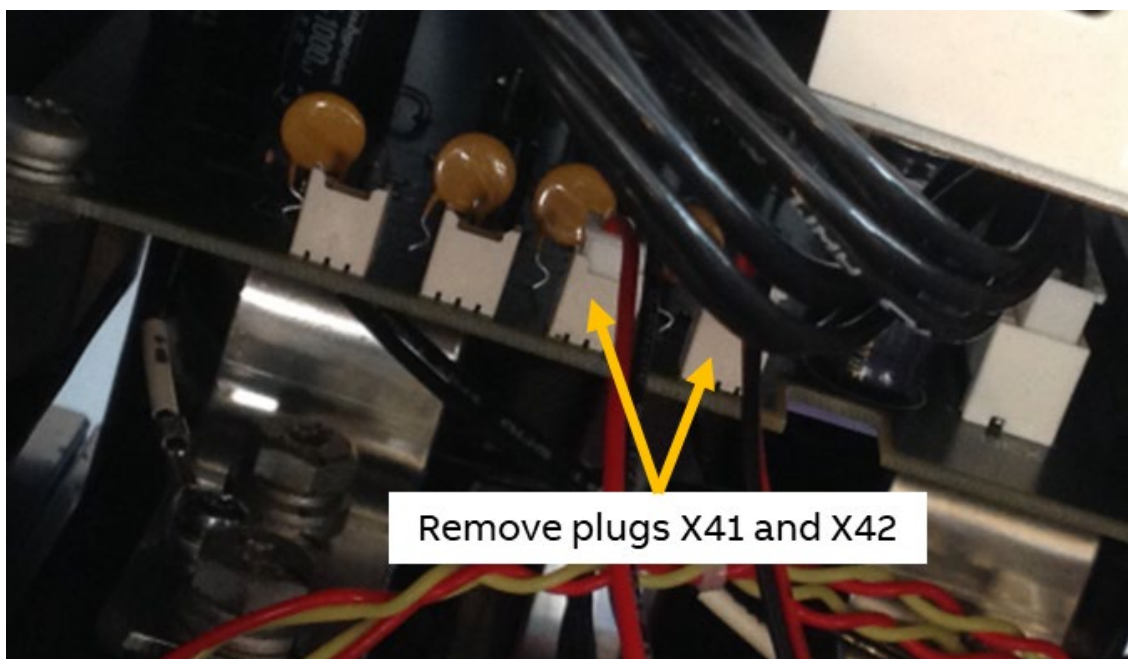
2. Remove the field connection plug X10 and the two fan arrangement screws.



3. Move the fan arrangement up and then forward.



4. Remove plugs X41 and X42 on the SDCS-PIN-K02.



5. Remove the fan holding screws of the broken fan.



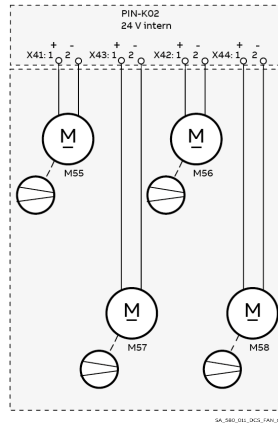
6. Attach the new fan, reassemble all parts, and check for correct blow direction of each fan. The air should be sucked out of the module.

Exchange cooling fans size K3 (four fans)



Before the work starts, disconnect the drive completely from the power supply, then check the voltage free condition and make sure everything is located in an electrical and mechanical safe condition!

Drives size DCS580-S0b-0470-05 ... DCS580-S0b-0520-05 and using four 24 V_{DC} fans.

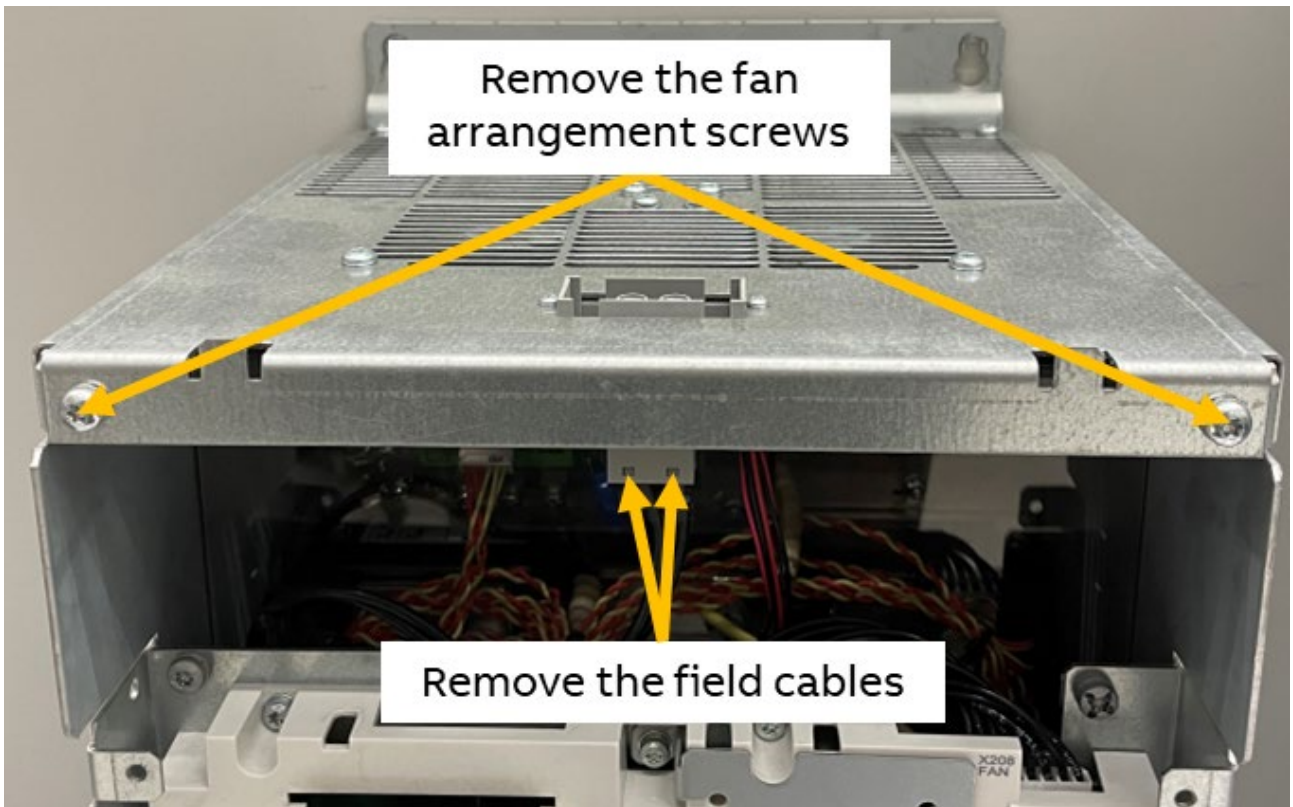


Removing the cooling fan

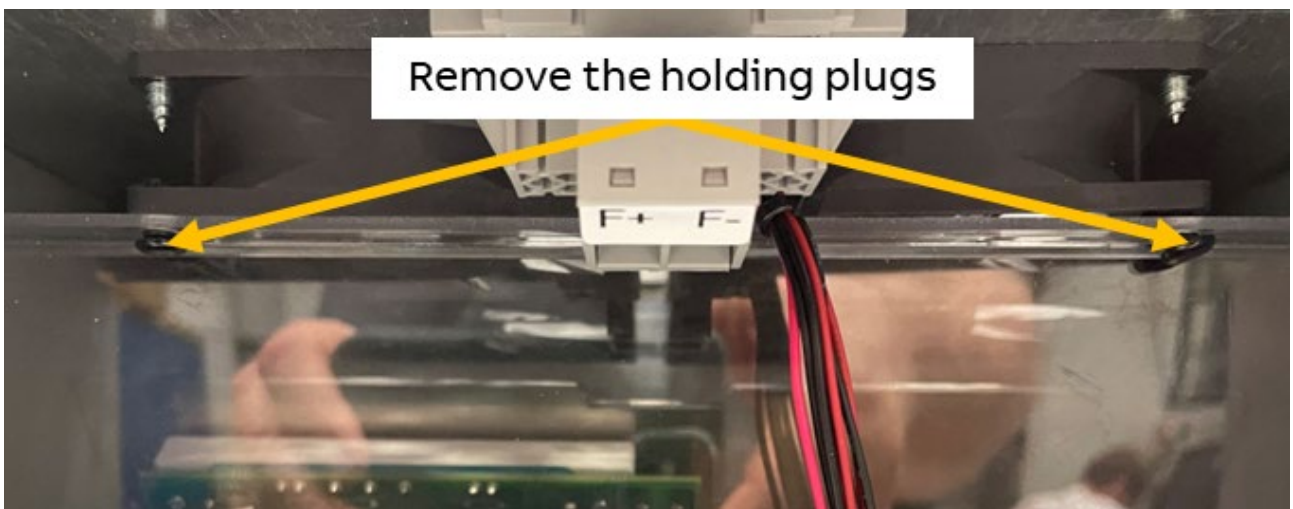
1. Remove the control panel and the design cover.



2. Remove the two fan arrangement screws and the field cables.
Note: Mark the field cables.



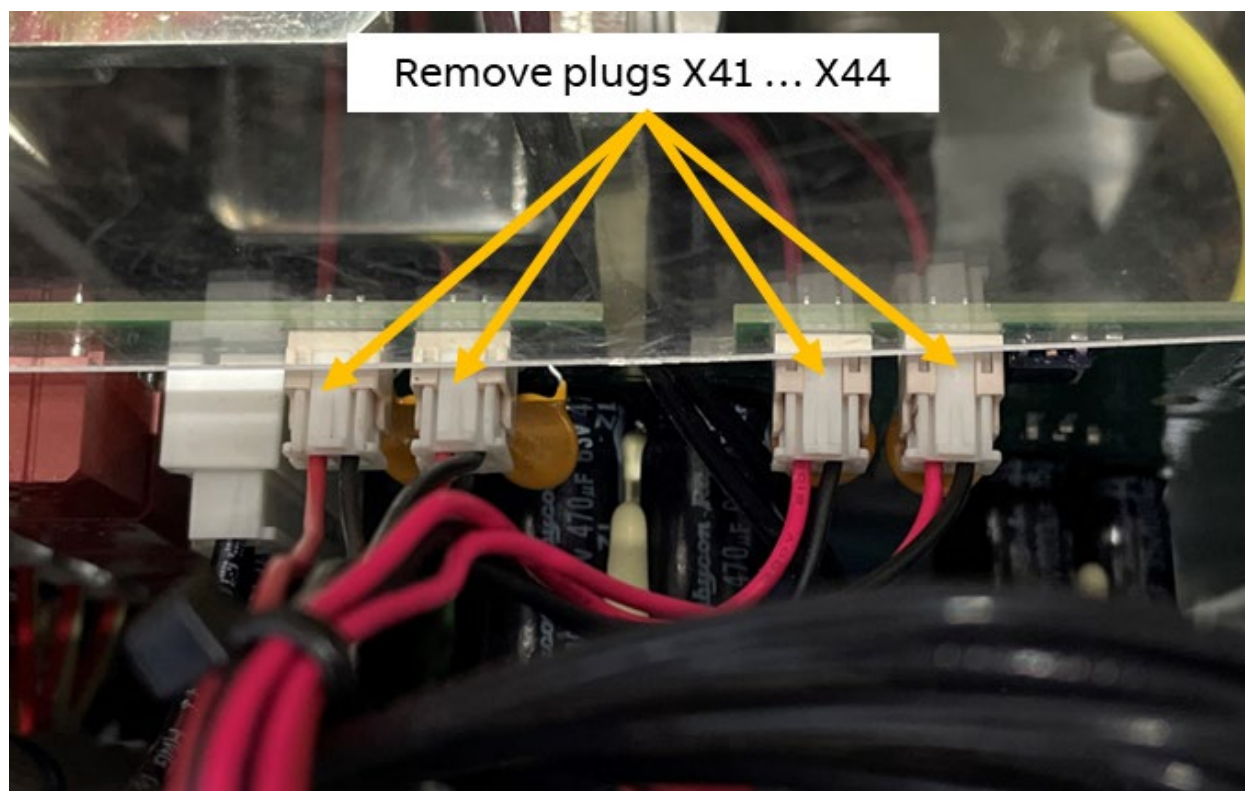
3. Remove the holding plugs by pulling out the head (use a flat screwdriver).



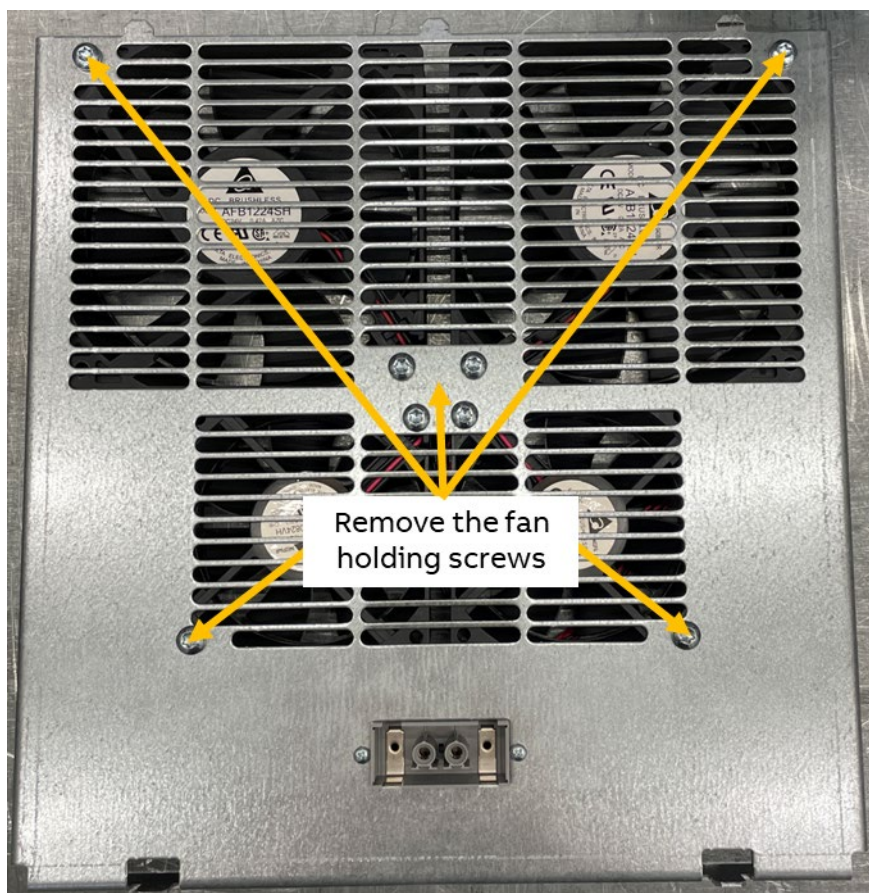
4. Move the fan arrangement forward and up.



5. Remove plugs X41 ... X44 on the SDCS-PIN-K02.



6. Remove the fan holding screws of the broken fan.



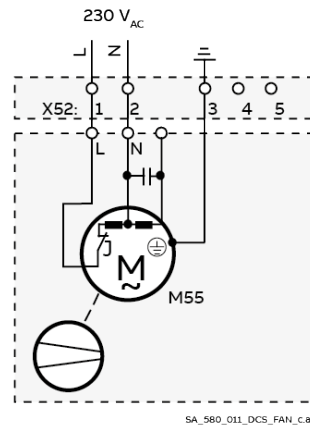
7. Attach the new fan, reassemble all parts, and check for correct blow direction of each fan. The air should be sucked out of the module.

Exchange cooling fan size K4



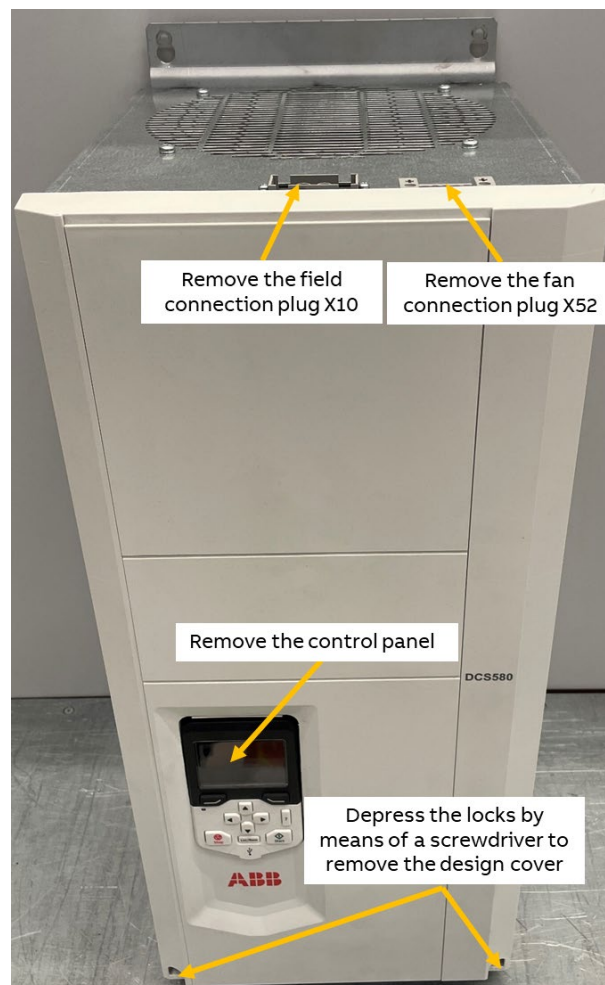
Before the work starts, disconnect the drive completely from the power supply, then check the voltage free condition and make sure everything is located in an electrical and mechanical safe condition!

Drives size DCS580-S0b-0610-05 ... DCS580-S0b-1000-05 using one 230 V_{AC} fan.

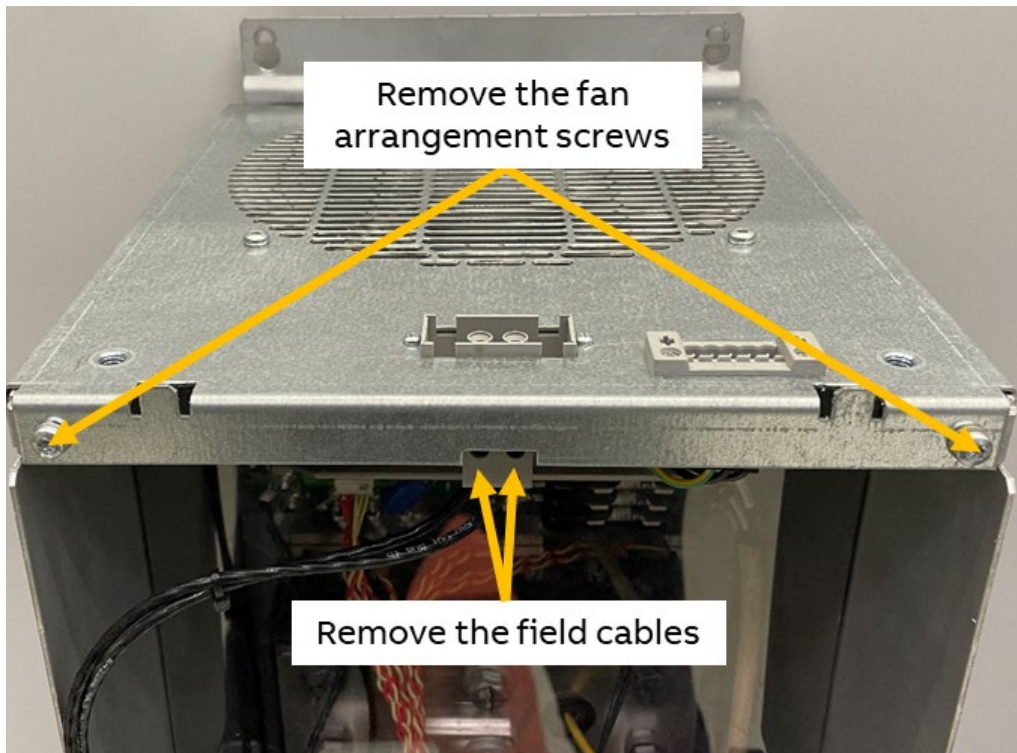


Removing the cooling fan

1. Remove the control panel, the design cover, and the field/fan connection plugs X10/X52 on the top of the drive housing.



2. Remove the two fan arrangement screws and the field cables.
Note: Mark the field cables.



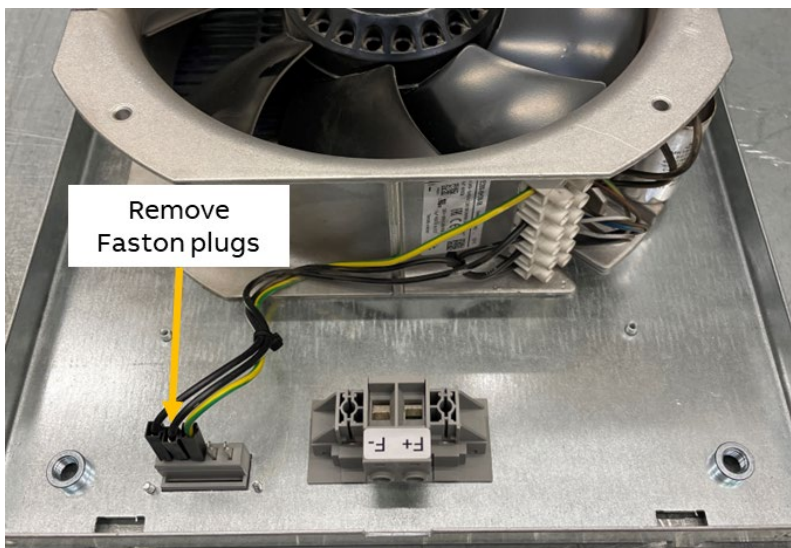
3. Move the fan arrangement forward and up.



4. Remove the fan holding screws.



5. Remove the Faston plugs from X52 and the broken fan.



6. Attach the new fan, reassemble all parts, and check for correct blow direction of the fan. The air should be sucked out of the module.

DCS580 firmware download

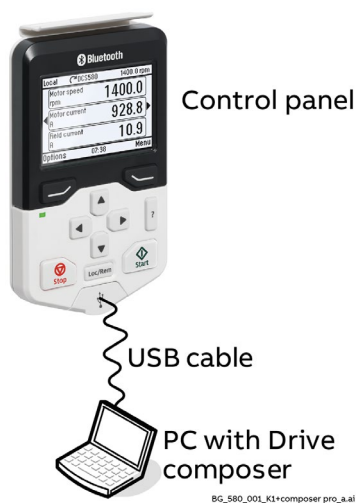
General

This file describes how to download firmware into the SDCS-CON-K02L.

Download SDCS-CON-K02L firmware using Drive Composer pro 2.9.1 or higher

Prepare firmware download

- Firmware loading package:
 - The firmware loading package can be found in section Firmware of the [DCS580 sales toolbox](#).
- Parameters:
 - The firmware download is setting all parameters back to default. Make a parameter backup beforehand using either the control panel or Drive Composer pro. This is for safety reasons only.
 - Nevertheless, Drive Composer pro will save the parameters (if activated) and write them back after the firmware download.
- Adaptive Program:
 - The firmware download deletes the Adaptive Program. In case an Adaptive Program is included, make a backup beforehand using Drive Composer pro. This is for safety reasons only.
 - Nevertheless, Drive Composer pro will save the Adaptive Program (automatically) and write it back after the firmware download.
- Application program:
 - The firmware download deletes the application program.
 - Make sure that a loading package of the application program is available.
 - The loading package of the application program can be downloaded the same way as a firmware loading package.
- The firmware download is done via a USB port and the control panel.



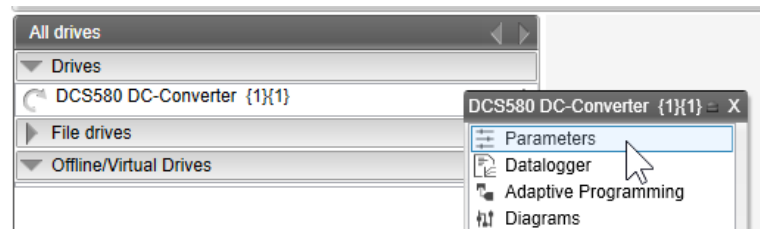
Firmware download

Make sure:

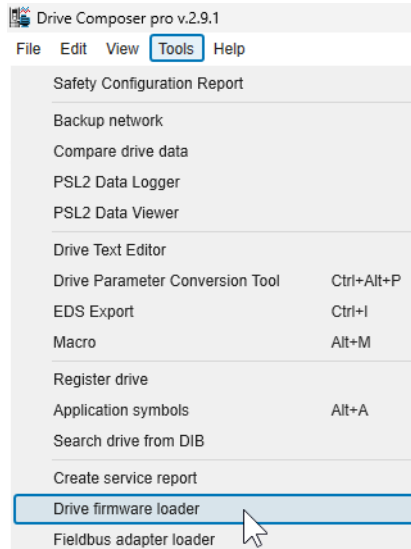
- The DCS580 is done with booting (energized, auxiliary voltage is on).
- The control panel is only connected to **one** DCS580.
- Open Drive Composer pro 2.9.1 or higher double click.



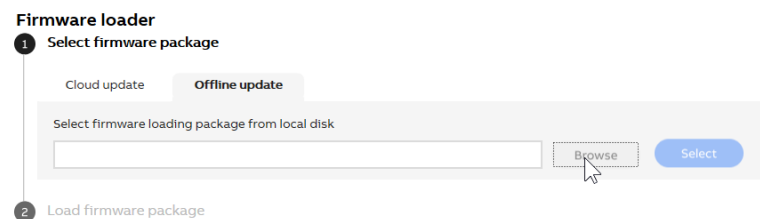
- Open the parameters.



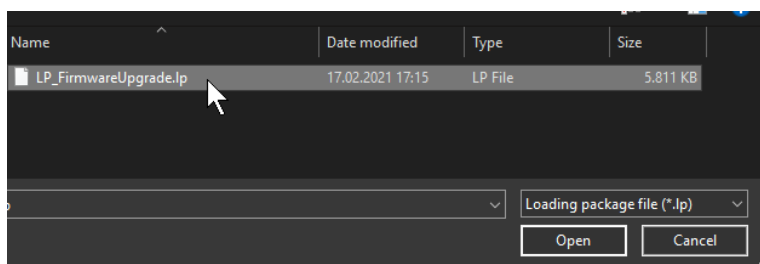
- Click *Tools - Drive firmware loader*.



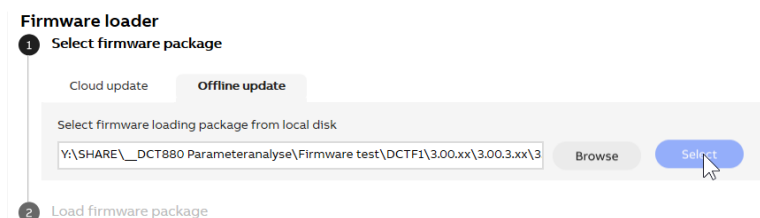
- Click *Browse* and go to the folder containing the desired firmware loading package.



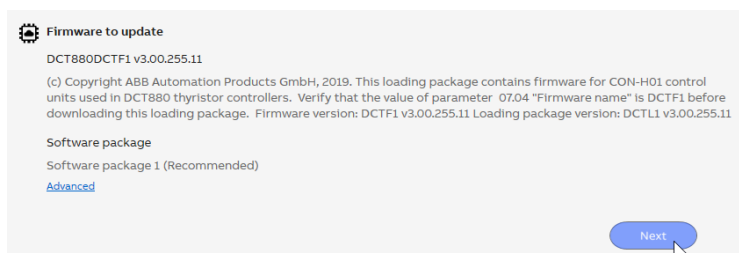
- Mark the desired firmware loading package and click *Open*.



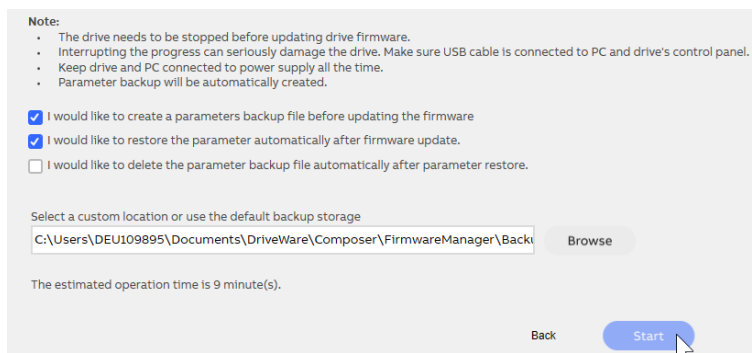
- Click *Select*.



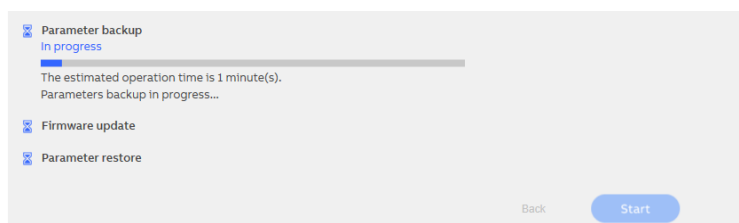
- Scroll down and click *Next*.



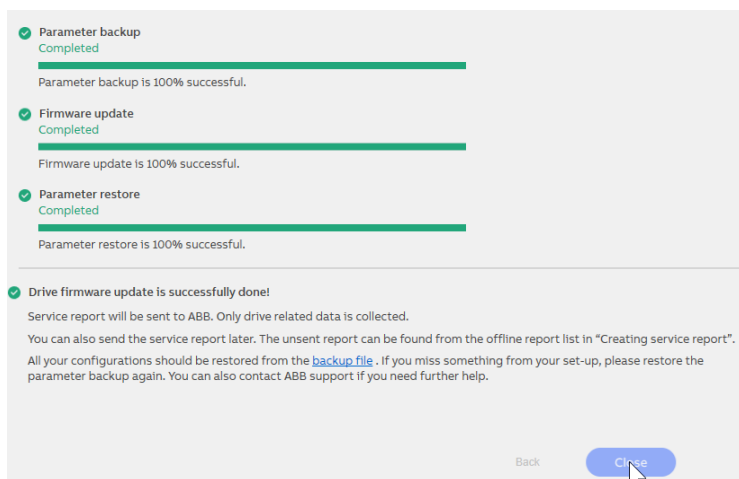
- Follow the hints under **Note:** at all times.
- Make sure to tick *I would like to ...*. This will save the parameters of the drive.
- Click *Start*.



- Status information.
 - *Parameter backup.*
 - *Firmware update.*
 - *Parameter restore.*



- Click *Close*.



Set type code

The type code, see 95.25 Set: Type code, is preset in the factory and is write protected. It identifies the drives current-, voltage-, temperature measurement and its quadrant type. The type code can only be changed as a whole or individually.

To change the type code as a whole, use the control panel and follow the instructions below:

- Un-protect the type code by means of 95.24 Service mode = Set: Type code.
- 95.25 Set: Type code = S01-0020-05 ... S02-1000-05 (details, see table below).

The drive's basic type code: DCS580-aab-cccc-ddef			
Product family:	DCS580		
Product type:	aa	= S0	Standard converter module
Bridge type:	b	= 1	Single bridge (2-Q)
		= 2	2 anti-parallel bridges (4-Q)
Module type:	cccc	=	Rated DC current (IP00)
Rated AC voltage:	dd	= 05	230 V _{AC} ... 525 V _{AC}
Power connection:	e	= X	Standard K1 ... K4
Revision code:	f	= 0	1 st generation

Attention: When using K1 ... K5 modules the current and voltage range of the type code setting is limited to max 1000 A_{DC} and max 500 V_{AC}.

- The change of the type code is immediately taken over and 95.24 Service mode must be set back to Normal mode by the user.
- The new values can be seen in group 7:
 - 07.60 Drive size.
 - 07.61 Drive block bridge 2 set.
 - 07.62 Drive DC current scaling set.
 - 07.64 Drive AC voltage scaling set.
 - 07.65 Drive max bridge temperature set.

To change the type code individually use the control panel and follow the instructions below:

- Un-protect the type code by means of 95.24 Service mode = Set: Type code.
- Set the type code individually by means of:
 - 95.26 Set: Drive block bridge 2.
 - 95.27 Set: Drive DC current scaling.
 - 95.28 Set: Drive AC voltage scaling.
 - 95.29 Set: Drive max bridge temperature.

The change of the type code is immediately taken over and 95.24 Service mode must be set back to Normal mode by the user.

- The new values can be seen in group 7:
 - 07.61 Drive block bridge 2 set.
 - 07.62 Drive DC current scaling set.
 - 07.64 Drive AC voltage scaling set.
 - 07.65 Drive max bridge temperature set.

DC-Motor neutral zone adjustment

Types concerned

DC-Motors.

Summary

Procedure to adjust the neutral zone of a DC-motor.

General

- Loosen the bolts holding the brush bridge.
- Connect a scope or a voltmeter between two adjacent brush-holder assemblies (holders with different polarity).
- Set the scaling on the scope or voltmeter to mV AC.
- Disconnect the excitation cables and connect 110 ... 240 V_{AC} between F+ and F- (a normal extension cord is sufficient).
- Start moving the brush bridge slowly in one direction and watch the scope or voltmeter. If the voltage increases move the brush bridge in the other direction.
- Try to get the voltage down to less than 50 mV AC. The smaller the better.
- When this is achieved, start tightening the brush bridge bolts and watch the voltage. Sometimes you have to do a little offset to compensate for movements during tightening.
- Finally, mark the correct neutral zone position in case the motor will be taken apart in the future.

Preventive Maintenance



WARNING

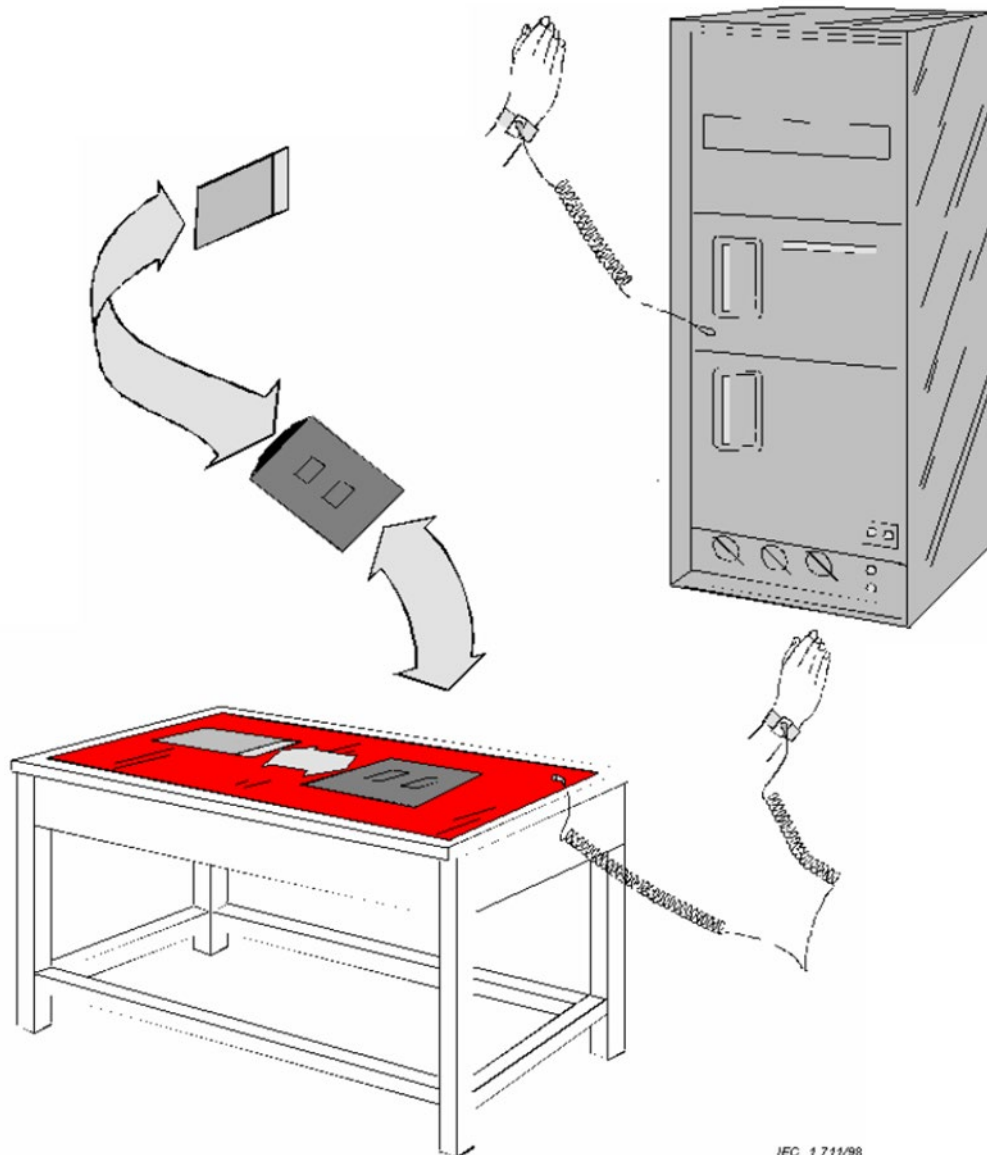
Before performing any maintenance, the chapter [Safety Instructions](#) at the beginning of this manual must be followed. Negligence of these instructions can cause injury or death.



WARNING

The printed circuit boards contain components sensitive to electrostatic discharge. Use an [ESD-field service kit](#) when handling the boards. Do not touch the boards unnecessarily.

Handling the boards:

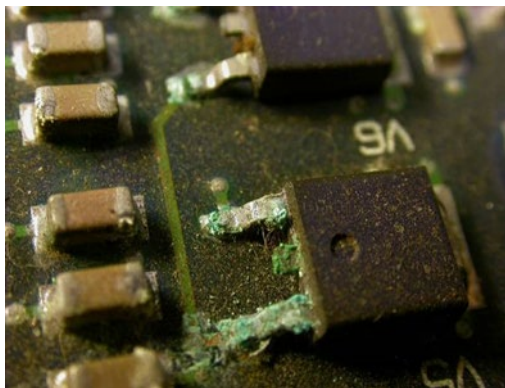


IEC 1711/98

When ESD sensitive boards are removed from the drive and transferred to a temporary work surface, they should be placed in a protective packaging. Handle unprotected ESD sensitive boards only when connected to the system and always place them in a protective package that is sealed. It is easiest to use the spare parts ESD package. Place the protective package on the ESD mat before opening.

Recommended regular maintenance

The DCS580 requires little maintenance if installed in an appropriate environment. Regular inspection according to the maintenance schedule is strongly recommended. Preventive maintenance prevents unexpected production stops and production loss. It also increases the availability of the drive. The environmental and operating conditions of the drive are also to be considered.



A harsh environment, such as high ambient temperature, humidity, dust, and cyclic heavy load, not only shortens the components lifetime but also the preventive maintenance and replacement intervals.

Maintenance schedule

	Years from start-up																					
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Start-up	P																					
Cooling																						
Air Cooled Unit:																						
Cooling fan DCS580 K1 (> 25 A) ... K4		I	I	I	I	I	R	I	I	I	I	I	R	I	I	I	I	I	R	I	I	I
Aging																						
DCS580 Power interface board SDCS-PIN-K02							(R)			R			(R)						R			
Connections & Surroundings																						
Flat cables							(R)			R			(R)						R			
Tightness of terminals				I			I			I			I				I		I			I
Door filters		I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I
Condition of contactors				I			I			I			I				I		I			I
Dustiness, corrosion and temperature		I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I
Quality of the mains voltage		I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I
Improvements																						
Based on product notes		I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I
Measurements																						
Basic measurements with the mains voltage		P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P	P
Spare Parts																						
Spare Parts		I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I	I

The service intervals and component replacements are based on the operational environment specified by ABB.

Legend:

- R** = Replacement of component
- I** = Inspection (visual inspection, correction and replacement if needed)
- P** = Performance of on-site work (commissioning, tests, measurements, etc.)
- (R)** = Replacement if high ambient temperature or cyclic heavy duty

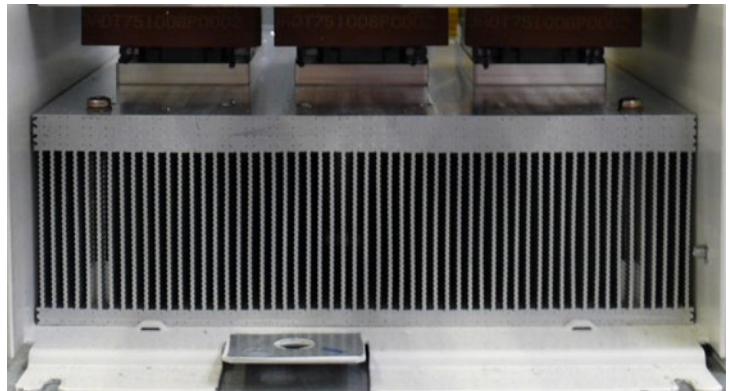
Annual preventive maintenance

The following actions have to take place:

- Check the cooling fans of drives size K1 ... K4.
The lifetime of the cooling fan is about 40,000 ... 100,000 hours depending on the drive type, the usage of the DCS580 and the environmental conditions (e.g., temperature). Fan failure can be predicted by means of increasing noise from its bearings and a gradual rise in heatsink temperature. If the DCS580 operates in a critical part of the process it is recommended to replace the fan once the above-mentioned symptoms appear.
- Check the door air filters.
- Clocked up filters prevent proper cooling of the drive. Wash dirty filters with water (60°C) and detergent.



- Check for dust, corrosion, and temperature inside the module/cabinet. The drive will run into overtemperature faults if the heatsinks are not clean.
- Pre-clean the heatsinks using an [ESD vacuum cleaner](#) then use compressed air to remove the dust from the heatsinks (the airflow must be from bottom to top). Fan rotation caused by the compressed air must be stopped in order to prevent damage.
- Use an [ESD vacuum cleaner](#) to clean the dust from the air inlet, air outlet, the interior of the cabinet and the electronic boards.
- Any signs of corrosion, especially at ground components, must be removed.

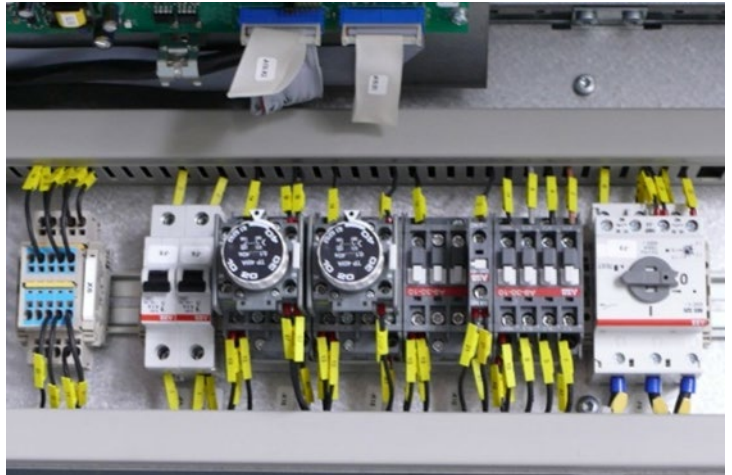


- Check the quality of the mains voltage (e.g., deviation from nominal voltage value, ...).

3 years preventive maintenance

The following additional actions have to take place:

- Check the tightness of all terminals.
- All connections should be inspected and checked for tightness.
- Check the condition of the contactors and relays.
- Contactors and relays should be checked for proper function.
- Check the fiber optic cables and their connections.



6 years preventive maintenance

The following additional actions have to take place:

- Replace the cooling fans of drives size K1 ... K4.
 - [Exchange the cooling fan size K1 \(one fan\).](#)
 - [Exchange the cooling fan sizes K2 ... K3 \(two fans\).](#)
 - [Exchange the cooling fan size K3 \(four fans\).](#)
 - [Exchange the cooling fan size K4.](#)

The manufacturer's estimation for the operational time of the classic cooling fans is:

- 3110UL: 70,000 hours at 25°.
- AFB122: 100,000 hours at 25°.
- W2E200: 45,000 hours at 60°C.
- W2E250: 40,000 hours.

The mayor problem is the failure of the bearings. The cooling fans are used in:

DCS580-S0b-0045-05 ... DCS580-S0b-0100-05	K1	1 x 3110UL 24 V internal
DCS580-S0b-0135-05 ... DCS580-S0b-0300-05	K2	2 x AFB122 24 V internal
DCS580-S0b-0315-05 ... DCS580-S0b-0450-05	K3	
DCS580-S0b-0470-05 ... DCS580-S0b-0520-05		2 x 3110UL 2 x AFB122 24 V internal
DCS580-S0b-0610-05 ... DCS580-S0b-0820-05	K4	1 x W2E200 230 V
DCS580-S0b-0900-05 ... DCS580-S0b-1000-05		1 x W2E250 230 V

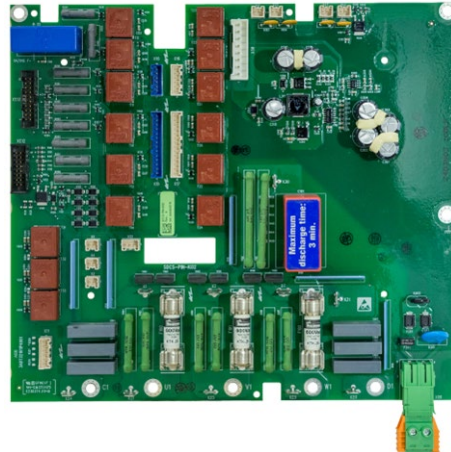
Aging of the components result in:

- Increased vibration due to the imbalance of the cooling fan (can last for several months).
- Increased temperature due to the stopped/too slow cooling fan (drive trips with overtemperature).
- Check the flat cables and their connections.

9 years preventive maintenance

The following additional actions have to take place:

- Replace the power interface board SDCS-PIN-K02 of drives size K1 ... K4 (DCS580-S0b-0020-05 ... DCS580-S0b-1000-05).
- The SDCS-PIN-K02 is located between the heat sink and the electronic tray. It operates continuously because it supplies the drive electronics even when the armature current is switched off. Therefore, all its components are exposed to hot conditions.
- Additionally, the electrolytic capacitors on the SDCS-PIN-K02 are aging. The SDCS-PIN-K02 is also equipped with several capacitors, which are sensitive to high ambient temperature.
- Aging of the components result in:
 - Damage other devices.
 - Trip the drive and cause breakdown time.
- Replace the flat cables of all drives size K1 ... K4.
- Environmental conditions, especially temperature and humidity could cause corrosion on the contact surfaces and embitterment of the insulation.
- Aging of the flat cables result in:
 - Contact problems.
 - Flat cable insulation breaks.



Preventive maintenance checklist for DCS580

Type of drive:	
Serial number:	
Last inspection:	

Drive size:	
Year of initial startup:	
Drive name:	

	Inspection cycle	K1 ... K4
1. Environment		
1.1 Checking the environment.	yearly	
1.2 Documentation checked and available.	yearly	
1.3 Checking the spare parts.	yearly	
2. Maintenance with no voltage applied		
2.1 Cleaning with ESD vacuum cleaner and/or soft brushes:		
Drive and cabinet.	yearly	
Air inlet and outlet filters (replace/clean if IP54 or if needed).	yearly	
Fins of fan.	yearly	
Heat sinks.	yearly	
2.2 Relays & connections:		
Inspect relays/contacts for proper functionality.	3 years	
Inspect electrical connections for tightness.	3 years	
Inspect for proper grounding.	yearly	
Inspect for corrosion.	yearly	
Inspect tightness of heatsink terminals (25 Nm required).	yearly	
Inspect connection of flat cables.	6 years	
3. Maintenance with mains voltage applied		
Create parameter backup.	yearly	
Check the level of all connected voltages.	yearly	
Inspect the condition of fans (check that all fans are operational).	yearly	

	Inspection cycle	K1 ... K4
4. Preventive Replacements		
4.1 Flat cables:		
SDCS-CON-K02L ... SDCS-PIN-K02 (XC12, XS13, X38).	9 years	
4.2 Boards:		
SDCS-PIN-K02.	9 years	
4.3 Fan		
Cooling fan drives size K1 ... K4.	6 years	

Inspect = visual inspection, correction and replacement if needed.

Remarks:

Date of inspection:	
Name of field service engineer:	
Sign of field service engineer:	

Appendix: Spare parts

For spare parts information please contact:

sales.productsupport@fi.abb.com.

Revision

Revision	Date	Comments
A	09.2025	First edition.
B		
C		
D		
E		

DCS Family



DCS580 modules The compact drive for machinery application

20 ... 1,000 A_{DC}
0 ... 610 V_{DC}
230 ... 525 V_{AC}
IP00

- Compact
- Robust design
- Adaptive Programming
- High field exciter current
- Extruder macro



DCS880 modules For safe productivity

20 ... 5,200 A_{DC}
0 ... 1,500 V_{DC}
230 ... 1,200 V_{AC}
IP00

- Safe torque off (STO) built in as standard
- Compact and robust
- Single drives, 20 A_{DC} to 5,200 A_{DC}, up to 1,500 V_{DC}
- IEC 61131 programmable
- Intuitive control panel and PC tool with USB connection and start up assistant
- Wide range of options to serve any DC motor application



DCS880-A cabinets Complete drive solutions

20 ... 20,000 A_{DC}
0 ... 1,500 V_{DC}
230 ... 1,200 V_{AC}
IP21 ... IP54

- Suitable for motoric and non motoric applications (e.g. electrolysis & hydrogen production)
- Individually adaptable to customer requirements
- User-defined accessories like external PLC or automation systems can be included
- High power solutions in 6- and 12-pulse up to 20,000 A_{DC}, 1,500 V_{DC}
- In accordance to usual standards
- Individually factory load tested
- Detailed documentation



DCT880 modules Thyristor power controller

20 ... 4,200 A_{AC}
110 ... 990 V_{AC}
IP00

- Precise power control in industrial heating applications
- Two or three phase devices
- Power optimizer for peak load reduction
- Built on ABB's all-compatible drives architecture
- Intuitive control panel and PC tool with USB connection and start up assistant
- Application control programs and drive application programming with IEC 61131 programming



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